A faster algorithm for three-tensor isomorphism over finite fields of odd characteristic

Abstract

The three-tensor isomorphism problem asks whether two three-tensors are related by invertible linear transformations in their three factors. We give a deterministic algorithm for this problem over a finite field $\mathbb{F}_q$ of odd characteristic. For tensors of length ℓ, the running time is $q^{O(\ell^{4/3}\log(\ell+2))}\operatorname{poly}(\log q)$, and an isomorphism is returned when one exists. The algorithm selects a short sequence of contractions and considers their adjoint algebra. A bound on the change of its simple factors yields subspaces on which the tensor has small analytic rank. The remaining quotient carries an action whose simple algebra factors have bounded dimension. We solve the resulting isomorphism problem by combining a constrained low analytic rank algorithm with an affine orbit algorithm. We prove the required structural bounds and the reductions between these problems.

1 Introduction

Let $\mathbb{F}_q$ be a finite field. A three-tensor of format $n_1 \times n_2 \times n_3$ is an array in $\mathrm{T}(n_1 \times n_2 \times n_3, \mathbb{F}_q)$. Its *length* is $\ell = n_1 + n_2 + n_3$. We regard it as a trilinear form on three ordered vector spaces. Two such tensors T and T' are isomorphic if there are invertible linear maps $g_i : V_i \to V_i'$ such that

$$T'(g_1 v_1, g_2 v_2, g_3 v_3) = T(v_1, v_2, v_3) \quad \text{for all } v_i \in V_i. \tag{1.1}$$

The three factors are kept in their given order.

Ivanyos, Mendoza, Qiao, Sun, and Zhang [IMQSZ24] gave an algorithm for three-tensor isomorphism with running time $q^{\widetilde{O}(\ell^{3/2})}$. Their algorithm first chooses left-right restrictions whose kernel is a matrix space of bounded maximum rank. A canonical shrunk subspace then gives a semi-canonical tensor with a zero block. Enumerating the boundary maps reduces its equivalence problem to conditioned alternating matrix-tuple congruence. The low analytic rank algorithm of [LR26] reduces the cost of completing an isomorphism from a zero block. To improve the bound for arbitrary tensors, we replace the first selection step and retain an additional orbit problem on the remaining quotient. We obtain the following bound.

Theorem 1.1. *Let q be odd. Given $T, T' \in \mathrm{T}(n_1 \times n_2 \times n_3, \mathbb{F}_q)$, one can decide deterministically whether T and T' are isomorphic, and find an isomorphism when one exists, in*

$$q^{O(\ell^{4/3}\log(\ell+2))}\operatorname{poly}(\log q)$$

bit operations, where $\ell = n_1 + n_2 + n_3$. The implicit constant is absolute.

In particular, when $n_i \le n$, the running time is $q^{O(n^c)}\operatorname{poly}(\log q)$ for every fixed $c > 4/3$. One may take $c = 7/5 < 3/2$.

Overview of the proof. For a sequence of contractions of T, let E be their adjoint algebra on a faithful module W of dimension $N = 2\ell$. Fix an integer $r \geq 1$. A simple factor of $E/J(E)$ has the form $\mathrm{M}(a, \mathbb{F}_{q^f})$ and has dimension fa^2 over $\mathbb{F}_q$. We distinguish the factors with $fa^2 > 100r^2$. The two-sided ideal generated by these factors determines a subspace $W_{10r}(E)$. Both the ideal and the subspace are independent of the lifts used to define them.

We use a potential that records the dimensions of the simple factors, weighted by their multiplicities in W. After $O(N \log(N+2)/r)$ contractions have been selected, every additional contraction changes the potential by less than r and decreases $\dim W_{10r}(E)$ by less than r. We prove that this implies an analytic rank bound

$$r + \frac{N^2}{100r^2}$$

for the contractions restricted to $W_{10r}(E)$. The proof treats the radical of E together with its semisimple quotient. In particular, it controls the blocks joining different composition factors of W.

The intersections with the three original factors give subspaces $A_i \leq V_i$. The part of T involving at least one A_i can be encoded by a tensor of analytic rank at most $3r + 3N^2/(100r^2)$. We solve its isomorphism problem together with the selected contractions and the subspace constraints. This gives a transporter and generators of its complete stabilizer. On the remaining quotient factors V_i/A_i, this group acts through algebra unit groups whose simple factors have dimension at most $100r^2$. A composition series reduces the remaining affine orbit problem to finite linear images and additive groups over the prime field.

The images of the selected source vectors are enumerated in the target. Thus the total running time is

$$q^{O(N^2 \log(N+2)/r + N + r^2 + N^4/r^4)} \operatorname{poly}(\log q).$$

Taking $r = \lceil N^{2/3} \rceil$ proves Theorem 1.1. The low analytic rank algorithm, the affine orbit algorithm, and the algebraic bound are proved below before they are applied to the input tensors.

2 Preliminary

Notations. For a positive integer n, let $[n] = \{1, \ldots, n\}$. All vector spaces, tensor products, and algebras are over $\mathbb{F} = \mathbb{F}_q$ unless specified otherwise. Vectors are columns, and the transpose of a matrix A is denoted by A^{t}. We write $\mathrm{M}(n_1 \times n_2, \mathbb{F})$ for the space of n_1 by n_2 matrices over $\mathbb{F}$, and $\mathrm{M}(n, \mathbb{F})$ when the matrices are square. We abbreviate $\mathrm{M}(n, \mathbb{F}_q)$ to $\mathrm{M}(n, q)$. Its group of invertible matrices is $\mathrm{GL}(n, q)$; for a vector space V, the corresponding group is $\mathrm{GL}(V)$. All logarithms have base 2, except for the logarithm in the definition of analytic rank. All running times count bit operations. The field is given explicitly, so its operations have cost polynomial in $\log q$.

Matrix spaces and congruence. A matrix space $\mathcal{A} \leq \mathrm{M}(n_1 \times n_2, \mathbb{F})$ is given by a basis. For $U \leq \mathbb{F}^{n_2}$, put $\mathcal{A}(U) = \langle Au : A \in \mathcal{A}, u \in U \rangle$. Its maximum rank is $\operatorname{mrk}(\mathcal{A}) = \max_{A \in \mathcal{A}} \operatorname{rank} A$. Its noncommutative rank is

$$\operatorname{ncrk}(\mathcal{A}) = \min_{U \leq \mathbb{F}^{n_2}} \big(n_2 - \dim U + \dim \mathcal{A}(U) \big).$$

This agrees with the usual noncommutative rank, with zero rows or columns added when necessary. An ordered tuple $\mathbf{A} = (A_1, \ldots, A_m)$ of square matrices is congruent to $\mathbf{B} = (B_1, \ldots, B_m)$ if $P^{\mathsf{t}} B_i P = A_i$ for one invertible P and every i. In odd characteristic a matrix is alternating if $A^{\mathsf{t}} = -A$. Congruence algorithms below allow degenerate alternating tuples.

Analytic rank. Fix a nontrivial additive character $\chi : \mathbb{F}_q \to \mathbb{C}^\times$. For a trilinear form $T : V_1 \times V_2 \times V_3 \to \mathbb{F}_q$, define

$$\mathrm{bias}(T) = \mathbb{E}_{v_1,v_2,v_3}\chi(T(v_1,v_2,v_3)), \qquad \mathrm{ark}(T) = -\log_q \mathrm{bias}(T).$$

Averaging over one variable shows that the bias is positive, real, and independent of χ. If T_x is the matrix obtained by fixing one variable, then

$$\mathrm{bias}(T) = \mathbb{E}_x q^{-\operatorname{rank} T_x}. \tag{2.1}$$

We also use this notation for a bilinear map $B : X \times Y \to Z$: its analytic rank means that of $(x, y, z^*) \mapsto z^*(B(x, y))$. Consequently $\mathrm{bias}(B) = \mathbb{E}_x q^{-\operatorname{rank} B_x}$.

Analytic rank is invariant under invertible changes in the factors. Restricting a factor to a subspace cannot increase analytic rank: in (2.1), average over a different factor and observe that each resulting matrix rank can only decrease. An arbitrary linear pullback is a restriction to its image followed by a surjection, and a surjection does not change the average. We shall also use the subadditivity theorem $\mathrm{ark}(S + T) \le \mathrm{ark}(S) + \mathrm{ark}(T)$ of Lovett [Lov19].

Algebras and modules. An algebra $E \le \mathrm{End}(W)$ contains the identity and is given by a vector space basis. An E-submodule is an E-invariant subspace of W. A composition series is an invariant flag whose successive nonzero quotients are simple modules, meaning that they have no proper nonzero invariant subspace. The radical $J(E)$ is the largest nilpotent two-sided ideal of E. The semisimple quotient is a product of matrix algebras over finite extensions of $\mathbb{F}_q$. We recall the structural statements needed for these algebras, and prove their additional consequences, in Section 6.

Groups and transporters. All groups acting on vector spaces are supplied by matrix generators. A transporter is the set of group elements taking one prescribed object to another. If it is nonempty and g_0 is one such element, it is g_0H, where H is the stabilizer of the source. Returning a transporter means returning g_0 and generators of H. A word in group generators may be stored as a straight-line program, whose instructions are multiplication and inversion; its length is the number of instructions. Such programs are evaluated in that length times the cost of a matrix operation.

3 Tensor isomorphism at low analytic rank

In this section the field $\mathbb{F} = \mathbb{F}_q$ has odd characteristic p. The three factors are ordered. Besides a trilinear form on $V_1 \times V_2 \times V_3$, we allow the following additional data: an ordered list of bilinear forms, each on a specified pair of distinct factors; an ordered list of linear forms; and an ordered list of subspaces of $V_1 \oplus V_2 \oplus V_3$. A triple $g = (g_1, g_2, g_3)$ transports these data if it transports each listed form and maps each listed subspace to its corresponding subspace. A linear form on the direct sum can equivalently be specified by its three factor components. The same convention applies to a sum of bilinear forms on different pairs of factors.

Theorem 3.1. *Let T, T' be trilinear forms over $\mathbb{F}$, with corresponding factor dimensions equal and at most n, where $n \ge 1$. Suppose that both inputs also contain additional data of the kinds just described. There is a deterministic algorithm which decides whether there is a triple g transporting all these data and satisfying $g^*T' = T$. If there is one, the algorithm returns such a triple g_0 and generators of the complete stabilizer H of the first input. The set of all solutions is then g_0H. The running time is*

$$q^{O(n+\mathrm{ark}(T)^2)}\,\mathrm{poly}(L, \log q),$$

where L is the encoded size of the two inputs. All constants and polynomial degrees are absolute.

The construction follows the low analytic rank algorithm in [LR26]. Lemma 3.2 uses the averaging argument of Lampert [Lam25, Lemmas 2.1–2.2]. Lemma 3.3 uses the rank-cover argument attributed to Haramaty and Shpilka in Briët, Christandl, Leigh, Shpilka, and Zuiddam [BCLSZ25, Lemma 6.6]. We give these proofs, the zero-block completion, and the reductions needed to retain all the prescribed data.

3.1 A contraction kernel and a canonical pair of subspaces

Write the factors as X, V, Z, put $U = X^*$, and express the tensor as

$$T(x, v, z) = x^{\mathrm{t}} A_z v, \qquad A_z \in \mathrm{Hom}(V, U).$$

The map $z \mapsto A_z$ is linear. Let $\mathcal{B} = \{A_z : z \in Z\}$. We retain the original spaces, including all directions on which T vanishes identically. For a matrix space $\mathcal{K} \leq \mathrm{Hom}(V, U)$, write

$$\mathrm{mrk}(\mathcal{K}) = \max_{A \in \mathcal{K}} \mathrm{rank}\, A, \qquad \mathcal{K}(W) = \langle Aw : A \in \mathcal{K}, w \in W \rangle.$$

We use the shrunk-subspace characterization of noncommutative rank:

$$\mathrm{ncrk}(\mathcal{K}) = \min_{W \leq V} \big(\mathrm{codim}_V W + \dim \mathcal{K}(W)\big). \tag{3.1}$$

This formula also applies to rectangular matrix spaces. If W attains the minimum, bases adapted to W and $\mathcal{K}(W)$ exhibit a zero block after deleting $\mathrm{codim}_V W$ columns and $\dim \mathcal{K}(W)$ rows.

For completeness, if χ is a nontrivial additive character of $\mathbb{F}$, averaging first over x and then over v gives

$$\mathrm{bias}(T) = \mathbb{E}_{z \in Z} q^{-\operatorname{rank} A_z} = \mathbb{E}_{A \in \mathcal{B}} q^{-\operatorname{rank} A} = \Pr_{A \in \mathcal{B},\ v \in V}[Av = 0]. \tag{3.2}$$

Here $\mathrm{bias}(T) = \mathbb{E}\chi(T)$ and $\mathrm{ark}(T) = -\log_q \mathrm{bias}(T)$. The middle equality holds because all fibers of the linear map $Z \to \mathcal{B}$ have the same cardinality.

Lemma 3.2. *Put $a = \mathrm{ark}(T)$. There is a vector $v_0 \in V$ such that*

$$\mathcal{K}_{v_0} = \{A \in \mathcal{B} : Av_0 = 0\}, \qquad b_{v_0} = \mathrm{codim}_{\mathcal{B}} \mathcal{K}_{v_0}$$

satisfy

$$b_{v_0} \leq a + 1, \qquad \mathrm{mrk}(\mathcal{K}_{v_0}) \leq 2\lfloor a + 2 \rfloor.$$

Proof. Let $m = \dim \mathcal{B}$, $v = \dim V$, and $\mathcal{Z} = \{(A, w) \in \mathcal{B} \times V : Aw = 0\}$. Equation (3.2) gives $|\mathcal{Z}| = q^{m+v-a}$. For the uniform distribution on $\mathcal{Z}$, counting first by A and then by w gives, respectively,

$$\mathbb{E}_{\mathcal{Z}} q^{\operatorname{rank} A} = q^a, \qquad \mathbb{E}_{\mathcal{Z}} q^{b_w} = q^a.$$

Conditioned on w, the matrix A is uniform in $\mathcal{K}_w$. Consequently some v_0 satisfies

$$q^{b_{v_0}} + \mathbb{E}_{A \in \mathcal{K}_{v_0}} q^{\operatorname{rank} A} \leq 2q^a \leq q^{a+1}.$$

It follows that $b_{v_0} \leq a + 1$ and, since the first summand is positive, that the second summand is strictly less than q^{a+1}. Markov's inequality therefore gives

$$\Pr_{A \in \mathcal{K}_{v_0}}[\operatorname{rank} A > a + 2] < q^{-1} \leq \tfrac{1}{2}.$$

Let $\mathcal{G}$ consist of the matrices of rank at most $\lfloor a + 2 \rfloor$ in $\mathcal{K}_{v_0}$. For every $B \in \mathcal{K}_{v_0}$, the sets $\mathcal{G}$ and $B + \mathcal{G}$ intersect, since each has more than half the elements of $\mathcal{K}_{v_0}$. Thus $B = A_1 - A_2$ for two matrices in $\mathcal{G}$, and the asserted rank bound follows. □

Lemma 3.3. *For every matrix space over any field,*

$$\operatorname{ncrk}(\mathcal{K}) \le 4 \operatorname{mrk}(\mathcal{K}).$$

Proof. Put $r = \operatorname{mrk}(\mathcal{K})$. The assertion is immediate when $r = 0$. Otherwise choose a rank-r member and change row and column bases to make it $A = \operatorname{diag}(I_r, 0)$, with the rectangular interpretation when necessary. Let $\overline{\mathcal{K}}$ be the lower-right restriction of $\mathcal{K}$.

Fix $B \in \mathcal{K}$, and let s be the rank of its lower-right block. Row and column operations inside that block put its nonzero pivots in an I_s block. Clear the remaining entries in these pivot rows and columns by using them. All these operations preserve A: the pivot rows and columns of A are zero. After reordering, the two matrices have the form

$$B = I_s \oplus D, \qquad A = 0_s \oplus A_0, \qquad \operatorname{rank} A_0 = r.$$

Since $B, A + B \in \mathcal{K}$,

$$s + \operatorname{rank} D \le r, \qquad s + \operatorname{rank}(A_0 + D) \le r.$$

The inequality $\operatorname{rank} A_0 \le \operatorname{rank} D + \operatorname{rank}(A_0 + D)$ now implies $s \le r/2$. Hence $\operatorname{mrk}(\overline{\mathcal{K}}) \le \lfloor r/2 \rfloor$. Deleting the first r rows and first r columns costs $2r$. Inductively the remaining space has a row-and-column cover of size at most $4\lfloor r/2 \rfloor$. The combined cover has size at most $4r$, and (3.1) gives the result. The construction in this proof takes place over the original field. □

For a fixed matrix space $\mathcal{K}$, define

$$\delta(W) = \dim W - \dim \mathcal{K}(W), \qquad \delta_* = \max_{W \le V} \delta(W) = \dim V - \operatorname{ncrk}(\mathcal{K}).$$

The function δ is supermodular. Indeed, $\mathcal{K}(W_1 + W_2) = \mathcal{K}(W_1) + \mathcal{K}(W_2)$ and $\mathcal{K}(W_1 \cap W_2) \subseteq \mathcal{K}(W_1) \cap \mathcal{K}(W_2)$ give

$$\delta(W_1) + \delta(W_2) \le \delta(W_1 + W_2) + \delta(W_1 \cap W_2).$$

Thus the maximizing subspaces are closed under sums and intersections. Their intersection is the unique smallest maximizer, denoted $W_*(\mathcal{K})$. Put

$$D_*(\mathcal{K}) = \mathcal{K}(W_*(\mathcal{K})), \qquad f = \operatorname{codim}_V W_*(\mathcal{K}), \qquad d = \dim D_*(\mathcal{K}). \tag{3.3}$$

Then $d + f = \operatorname{ncrk}(\mathcal{K})$. Uniqueness shows that, if $\mathcal{K}' = P\mathcal{K}S^{-1}$, then

$$W_*(\mathcal{K}') = SW_*(\mathcal{K}), \qquad D_*(\mathcal{K}') = PD_*(\mathcal{K}). \tag{3.4}$$

For rank computation, we use the deterministic algorithm of Ivanyos, Qiao, and Subrahmanyam [IQS18, Theorem 1.5 and Corollary 1.7(2)]. Given a basis for a matrix space over a finite field, it computes the rank in (3.1) in time polynomial in the dimensions and $\log q$, including for small finite fields. Rectangular spaces may be padded by zero rows or columns. The subspaces in (3.3) can be obtained using only these rank values, as follows. Enumerate the hyperplanes $J \le V$. Such a hyperplane contains $W_*(\mathcal{K})$ exactly when

$$\dim J - \operatorname{ncrk}(\mathcal{K}|_J) = \delta_*. \tag{3.5}$$

If it contains W_*, that subspace attains the right side in J. Conversely, equality gives a maximizing subspace inside J, which contains W_*. Intersecting all hyperplanes that pass this test gives W_*; the intersection of an empty collection is V. There are at most $q^{\dim V}$ hyperplanes. Thus the pair can be computed deterministically in $q^{O(\dim V)}$ times a polynomial in the input size and $\log q$.

For each $v_0 \in V$ define the full coefficient preimage

$$Z_0(v_0) = \{z \in Z : A_z v_0 = 0\}.$$

Its image is $\mathcal{K}_{v_0}$ and its codimension in Z is b_{v_0}. Use $W_*(\mathcal{K}_{v_0})$ as a subspace of V, and $D_*(\mathcal{K}_{v_0})^\perp$ as a subspace of X. The tensor vanishes on the product of these two subspaces and $Z_0(v_0)$. Lemmas 3.2 and 3.3 imply that some v_0 has the three codimensions d, f, b satisfying

$$b^2 + d^2 + f^2 \le (a+1)^2 + 64(a+2)^2, \qquad a = \operatorname{ark}(T). \tag{3.6}$$

In fact $b \le a+1$ and $d+f \le 8\lfloor a+2 \rfloor$. The vector minimizing $b^2+d^2+f^2$ is found by enumerating V and comparing these integer values.

3.2 Algorithms for ordered alternating forms

We use two algorithms for ordered tuples of alternating matrices over $\mathbb{F}$. Given two tuples (A_i) and (B_i) of the same length, the algorithm of Ivanyos–Qiao decides whether $G^{\mathsf{t}} A_i G = B_i$ for every i, and returns G when it exists. The deterministic version of their theorem has running time polynomial in the matrix dimension, tuple length, $\log q$, and p [IQ19, Theorem 7(1), arXiv version]. For one tuple, the algorithm of Brooksbank–Wilson returns generators of its full isometry group. Degenerate forms are allowed. Their Theorem 1.1 and the deterministic variant in Section 7.4 give a running time polynomial in the input size and p [BW12]. Field operations have polynomial bit complexity in $\log q$. We will reduce the following two tasks to these algorithms.

Lemma 3.4. *Let a vector space have a specified direct-sum decomposition. Congruence of ordered alternating tuples under the group preserving each summand can be decided, and a congruence can be found, in deterministic time polynomial in the dimensions, tuple length,* $\log q$*, and* p*. Generators of the full group preserving the summands and the tuple can be computed within the same bound.*

Proof. Write the space as $\mathbb{F}^s$, and let Π_j be its coordinate projections onto the specified summands. On three copies of $\mathbb{F}^s$, define

$$\Omega_{12}(P) = \begin{pmatrix} 0 & P & 0 \\ -P^{\mathsf{t}} & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \qquad \Omega_{13} = \begin{pmatrix} 0 & 0 & I \\ 0 & 0 & 0 \\ -I & 0 & 0 \end{pmatrix}.$$

Include the same forms $\Omega_{12}(I), \Omega_{13}$, and $\Omega_{12}(\Pi_j)$ in both tuples, and replace each input A_i by $\operatorname{diag}(A_i, 0, 0)$. The radicals of $\Omega_{12}(I)$ and Ω_{13} are respectively the third and second copies. Therefore every congruence of the enlarged tuples has the form

$$S = \begin{pmatrix} G & 0 & 0 \\ L & H & 0 \\ M & 0 & K \end{pmatrix}.$$

The two identity cross forms force $H = K = G^{-\mathsf{t}}$. The projector equations yield $G^{\mathsf{t}} \Pi_j G^{-\mathsf{t}} = \Pi_j$, which is equivalent to $G\Pi_j = \Pi_j G$. Thus G preserves all summands. The remaining input equations are exactly $G^{\mathsf{t}} A_i G = B_i$. Conversely every such G lifts to $\operatorname{diag}(G, G^{-\mathsf{t}}, G^{-\mathsf{t}})$. Ordinary tuple congruence consequently gives the desired decision and map. For one tuple, projection $S \mapsto G$ is a surjective group homomorphism onto the desired group, so projecting a full generating set gives a full generating set. Zero-dimensional summands can be omitted. □

An involution on an $\mathbb{F}$-algebra E is an $\mathbb{F}$-linear map $a \mapsto a^*$ satisfying $(ab)^* = b^*a^*$ and $(a^*)^* = a$. For a unital algebra with involution, write

$$U(E,*) = \{a \in E : a^*a = aa^* = 1\}.$$

An explicit right E-module is specified by its vector-space basis and the action matrices of a basis of E.

Lemma 3.5. *Let E be an explicit finite-dimensional unital $\mathbb{F}$-algebra with involution, and let M be an explicit unital right E-module. For $v, w \in M$, one can decide whether $va = w$ for some $a \in U(E,*)$ and return such an a. One can also compute generators of $\{a \in U(E,*) : va = v\}$. These tasks have deterministic running time polynomial in the supplied dimensions,* $\log q$, *and p.*

Proof. We give the full reduction to ordered alternating forms. All module actions in this proof are written on row vectors. Let $W = E_E \oplus M$, where E_E is the right regular module, and let $\rho(a)$ be the right-action matrix on W. On $V = W \oplus W^*$ define

$$R(a) = \operatorname{diag}\big(\rho(a), \rho(a^*)^{\mathsf{t}}\big), \qquad J = \begin{pmatrix} 0 & I \\ -I & 0 \end{pmatrix}.$$

The second summand here has the right action prescribed by the displayed formula. Both transpose and involution reverse products, so R is an algebra representation. It is faithful because W contains the regular module. Direct multiplication gives

$$R(a)^{\mathsf{t}} J = J R(a^*). \tag{3.7}$$

Consequently $R(a)$ is an isometry of J precisely when a is unitary. Indeed the isometry equation implies $a^*a = 1$ by faithfulness; in a finite-dimensional algebra a left inverse is also a right inverse.

Compute a basis $C_1, \dots, C_r$ of the commutant of $R(E)$ by linear algebra. Its commutant is exactly $R(E)$. To verify this, let B commute with every right E-module endomorphism of V. It commutes with projection onto the regular summand, and on that summand it commutes with all left multiplications. Hence it acts there as right multiplication by some $a \in E$. For every $v_0 \in V$, the module homomorphism $E_E \to V$, $x \mapsto v_0 x$, extends by zero on the other summands to an endomorphism of V. Commuting with this endomorphism and applying to $1 \in E_E$ gives $B(v_0) = v_0 a$. Thus $B = R(a)$, proving the claim for arbitrary E.

On $D = V \oplus V$ take the ordered alternating forms

$$A_L = \operatorname{diag}(J, 0), \quad A_R = \operatorname{diag}(0, J), \quad A_\times = \begin{pmatrix} 0 & I \\ -I & 0 \end{pmatrix}, \quad A_i = \begin{pmatrix} 0 & C_i^{\mathsf{t}} \\ -C_i & 0 \end{pmatrix} \quad (1 \le i \le r).$$

The radicals of A_L, A_R force every common isometry to be $\operatorname{diag}(X, Y)$. The equation for $A_\times$ gives $Y = X^{-\mathsf{t}}$, and the equations for A_i give $X^{-1} C_i X = C_i$ for all i. The double-commutant identity gives $X = R(a)$ for some $a \in E$. The equation for A_L, together with (3.7), is equivalent to $a \in U(E,*)$. The equation for A_R then follows as well: for the displayed standard J, $X^{\mathsf{t}} J X = J$ also implies $X^{-1} J X^{-\mathsf{t}} = J$. We have therefore obtained precisely the group

$$\big\{\operatorname{diag}\big(R(a), R(a)^{-\mathsf{t}}\big) : a \in U(E,*)\big\}. \tag{3.8}$$

Embed v, w as row vectors in the M summand of the first copy of V in D, and denote these rows by $\widehat{v}, \widehat{w}$. Add a three-dimensional space P with basis e_1, e_2, e_3. Extend all the old alternating forms

by zero on P. Also include, on P, each of the three elementary alternating forms on its coordinate pairs, and include the form

$$B_v = \begin{pmatrix} 0 & \widehat{v}^{\mathsf{t}} e_1^{\mathsf{t}} \\ -e_1 \widehat{v} & 0 \end{pmatrix}$$

on $D \oplus P$. In the second tuple replace B_v by B_w and leave all other forms the same. The old nondegenerate form $A_\times$ has radical P, so a congruence preserves P. The common radical of the three new forms is D, so it preserves D too. The three individual radicals inside P are its coordinate lines. An isometry on P is consequently diagonal, say with entries $\lambda_1, \lambda_2, \lambda_3$, and their pairwise products are all 1. It follows that all three entries are one scalar $\lambda \in \{1, -1\}$.

It follows that a congruence of these enlarged tuples is of the form $\mathrm{diag}(Z, \lambda I_3)$, where Z belongs to the group in (3.8), and the remaining equation is

$$\widehat{v} Z = \lambda \widehat{w}.$$

Multiplication of Z by λ gives the exact desired row correspondence. The matrix λZ still belongs to (3.8), since $\lambda 1_E$ is a central unitary element. Recover the resulting element of E from its matrix $R(a)$ by linear algebra. Ordinary tuple congruence thus decides and solves the module orbit problem.

For $w = v$, the map

$$\mathrm{diag}(Z, \lambda I_3) \longmapsto \lambda Z$$

is a surjective group homomorphism onto the stabilizer of $\widehat{v}$ in (3.8). Surjectivity follows by taking $\lambda = 1$ for every isometry fixing $\widehat{v}$. Computing the full isometry group of the enlarged tuple and projecting its generators therefore gives the complete module-vector stabilizer. Every space, commutant, and tuple used above has dimension polynomial in $\dim E + \dim M$. The algorithms stated at the beginning of this subsection prove the required deterministic bounds. □

3.3 Completion with a specified zero block

Suppose each original factor has been decomposed into a subspace and a chosen complement, and T vanishes on the product of the three subspaces. Use coordinates $(x, u), (y, v), (z, w)$, where x, y, z belong to the subspaces. The maps preserving them have the form

$$\begin{aligned} x &\longmapsto G_x x + H_x u, & u &\longmapsto D_u u, \\ y &\longmapsto G_y y + H_y v, & v &\longmapsto D_v v, \\ z &\longmapsto G_z z + H_z w, & w &\longmapsto D_w w. \end{aligned}$$

Fix invertible maps D_u, D_v, D_w on the three quotients. A known substitution then makes all three quotient maps identities. We will also allow any number of extra, individually specified variable spaces, with independent invertible maps. The cubic tensor uses only the original three factors. The extra spaces may occur in prescribed bilinear forms between distinct spaces and in prescribed linear forms. They will have no fixed quotient coordinates.

Lemma 3.6. *Once the three quotient maps have been fixed, equivalence of the cubic tensor and all the prescribed bilinear and linear forms under the displayed substitutions can be decided, and a substitution can be found, in deterministic time polynomial in the complete supplied dimensions,* $\log q$*, and* p*.*

Proof. Collect the original subspace coordinates and all the extra coordinates into a column t. Let $c = (u, v, w)^{\mathsf{t}}$ be the fixed quotient coordinates. The substitutions are

$$t \longmapsto Gt + Hc, \qquad c \longmapsto c, \tag{3.9}$$

where G preserves each specified variable space and H lies in the linear space $\mathcal{D}$ having only the blocks from u to x, from v to y, and from w to z. All rows of H corresponding to extra spaces are zero.

For each bilinear form B_j introduce a distinct fixed formal variable β_j. For each linear form L_k introduce a distinct fixed formal variable ζ_k. Form the homogeneous cubic polynomial

$$F = T + \sum_j \beta_j B_j + \sum_k \zeta_k^2 L_k.$$

Equality of formal coefficients expresses exactly the desired equalities of the tensor and all listed forms. Set $\omega = (c, \beta, \zeta)$. The degree of F in t is at most two, so it has a unique expression

$$F(t, \omega) = t^{\mathsf{t}} Q(\omega) t + \ell(\omega) t + \gamma(\omega), \tag{3.10}$$

where Q is symmetric and linear in ω, ℓ is a row of quadratic polynomials, and γ is cubic in ω. For Q, divide every cross coefficient equally between its two symmetric entries. This uses the assumption that p is odd. The equations are equations of formal polynomials over $\mathbb{F}$.

For this proof let F_1, F_2 be the two polynomials and seek a substitution with $F_1(Gt + Hc, \omega) = F_2(t, \omega)$. Substitution in (3.10) gives

$$Q_2 = G^{\mathsf{t}} Q_1 G, \tag{3.11}$$

$$\ell_2 = (\ell_1 + \Delta_H) G, \qquad \Delta_H = 2(Hc)^{\mathsf{t}} Q_1, \tag{3.12}$$

$$\gamma_2 = \gamma_1 + \ell_1 Hc + (Hc)^{\mathsf{t}} Q_1 (Hc). \tag{3.13}$$

Each coefficient matrix of Q_i is supported between one pair of distinct variable spaces. Replacing the lower of its two off-diagonal blocks by its negative makes the matrix alternating. Under the permitted block-diagonal G, its congruence equation is equivalent to the original symmetric one. Apply Lemma 3.4 to the ordered coefficient tuples. If they are inequivalent, reject. Otherwise apply the returned change of coordinates and henceforth write both leading terms as the same Q.

Let $Q_1, \ldots, Q_s$ now denote the coefficient matrices of this common polynomial matrix Q. Form the algebra of adjoint pairs

$$E = \{(r, s) : r, s \text{ preserve every specified variable space, } r^{\mathsf{t}} Q_i = Q_i s \text{ for all } i\}.$$

It is computed by linear equations. Multiplication and involution are

$$(r, s)(r', s') = (rr', s's), \qquad (r, s)^* = (s, r).$$

The symmetry of the Q_i ensures that this is a unital algebra with involution. Its unitary group consists exactly of

$$(G, G^{-1}), \qquad G^{\mathsf{t}} Q_i G = Q_i \quad \text{for every } i,$$

with the prescribed block decomposition.

Let $\mathcal{L}$ be the space of all row arrays of length $\dim t$ whose entries are homogeneous quadratic polynomials in ω. The first component of E gives the right action $\ell(r, s) = \ell r$ on $\mathcal{L}$. Define the linear map $\Delta : \mathcal{D} \to \mathcal{L}$ by $H \mapsto 2(Hc)^{\mathsf{t}} Q$, and let $\mathcal{U} = \Delta(\mathcal{D})$. Transposing the adjoint equations gives $Qr = s^{\mathsf{t}} Q$, whence

$$\Delta_H r = 2(Hc)^{\mathsf{t}} Qr = 2(sHc)^{\mathsf{t}} Q = \Delta_{sH}. \tag{3.14}$$

The matrix s preserves each variable space, so sH still belongs to $\mathcal{D}$. In particular its rows on all extra spaces are zero. Thus $\mathcal{U}$ is a right E-submodule, and $M = \mathcal{L}/\mathcal{U}$ is an explicit unital right module.

Lemma 3.5 decides whether there is a permitted G with $[\ell_1]G = [\ell_2]$ in M and finds one if it exists. Given G, solve the linear system

$$\Delta_H = \ell_2 G^{-1} - \ell_1, \qquad H \in \mathcal{D}.$$

It is soluble exactly when the two classes agree. Applying this substitution makes the actual Q and ℓ terms of the two polynomials equal.

Use Lemma 3.5 again to compute the full stabilizer of $[\ell]$. For each returned generator G_i, choose $H_i \in \mathcal{D}$ by solving

$$\Delta_{H_i} = \ell G_i^{-1} - \ell.$$

Also take an $\mathbb{F}_p$-basis of $\ker \Delta$ and include the substitutions (I, H) for those basis elements. These substitutions generate the full stabilizer S of the actual pair (Q, ℓ). Indeed its projection to G is exactly the stabilizer of $[\ell]$; every element of that stabilizer has a lift; and the kernel is the additive group $\ker \Delta$.

For $(G, H) \in S$, put

$$\eta(G, H) = \ell H c + (Hc)^{\mathsf{t}} Q (Hc).$$

The action on polynomials is a right action, with multiplication

$$(G, H)(G', H') = (GG', GH' + H).$$

Every substitution fixes ω, and it preserves Q and ℓ. Applying it to $t^{\mathsf{t}} Q t + \ell t$ therefore adds just $\eta(G, H)$. Composition shows

$$\eta(gg') = \eta(g) + \eta(g').$$

Its image is the $\mathbb{F}_p$-linear span of the increments of the computed generators of S. Prime-field linear algebra tests whether $\gamma_2 - \gamma_1$ belongs to that span. When it does, a product of generator powers gives the desired increment; compute the powers by repeated squaring. When it does not, this choice of quotient maps is impossible.

This constructs the required substitution. If $\dim t = 0$, only the fixed polynomial coefficients have to be compared. Otherwise, writing $r = \dim t$ and $h = \dim \omega$, we have $\dim E \le 2r^2$, $\dim \mathcal{L} = r\binom{h+1}{2}$, and at most $\binom{h+2}{3}$ coefficients in γ. All modules and tuples constructed from these spaces have dimensions bounded by fixed polynomials in $r + h$. The asserted deterministic bound follows from Lemmas 3.4 and 3.5, together with linear algebra over $\mathbb{F}$ and $\mathbb{F}_p$. □

3.4 Prescribed subspaces

We encode prescribed subspaces by additional bilinear forms on extra variable spaces of the kind allowed in Lemma 3.6. Suppose $S_j \le V_1 \oplus V_2 \oplus V_3$ and S_j' is its specified partner. Reject if their dimensions differ. Choose full-column-rank inclusion matrices

$$M_j : \mathbb{F}^{d_j} \longrightarrow V_1 \oplus V_2 \oplus V_3, \qquad M_j' : \mathbb{F}^{d_j} \longrightarrow V_1' \oplus V_2' \oplus V_3',$$

and write M_{ij}, M_{ij}' for their three row blocks.

For each original factor V_i introduce a separate space D_i of the same dimension, and prescribe the evaluation bilinear form with matrix I between D_i and V_i on both inputs. If their transformations are d_i and g_i, this form requires

$$d_i^{\mathsf{t}} g_i = I, \qquad \text{and hence } d_i = g_i^{-\mathsf{t}}.$$

For every j with $d_j > 0$, introduce a separate space P_j of dimension d_j, with transformation h_j. Prescribe the three bilinear forms between D_i and this same P_j having matrices M_{ij} and M'_{ij}. Their equations are

$$d_i^{\mathsf{t}} M'_{ij} h_j = M_{ij} \quad (1 \le i \le 3).$$

With the evaluation equations, these are equivalent to

$$M'_j h_j = \operatorname{diag}(g_1, g_2, g_3) M_j. \tag{3.15}$$

An invertible h_j satisfying this equation exists precisely when $gS_j = S'_j$. When it exists it is unique, since M'_j is injective. The d_i are unique as well. Therefore the solution set of the enlarged system projects bijectively onto the original solution set; for a stabilizer this projection is a group isomorphism.

Every added space is preserved individually. In the zero-block completion put it entirely among the coordinates t, with no quotient coordinates. The only permitted shear rows are still the three original ones. The evaluation and inclusion forms join the ordered bilinear list before the construction of Q, E, and M in Lemma 3.6. Every leading coefficient still joins distinct variable spaces, and (3.14) still holds because s preserves each of these spaces. In particular no extra nonzero row is introduced into sH. The inverse transpose $d_i = g_i^{-\mathsf{t}}$ is an arbitrary matrix inside the single space D_i, so it includes the entries determined by the original shear of g_i.

If there are k prescribed subspaces, the original and added spaces together have dimension at most $6n + 3nk$. There are at most $3k + 3$ new bilinear forms. The dimensions of $\mathcal{L}$, E, and the modules and tuples used above are fixed polynomials in these quantities and the number of prescribed forms. All extra spaces occur only in the polynomial-time procedures of Lemma 3.6.

3.5 The complete algorithm

We now prove the decision and search assertion of Theorem 3.1. Construct the original matrix space $\mathcal{B}$ on X, V, Z and perform the following operations.

1. For every $v_0 \in V$, compute $Z_0(v_0)$, $\mathcal{K}_{v_0}$, and its pair (W_*, D_*). Choose a vector minimizing the integer $b^2 + d^2 + f^2$ of (3.6), and retain its three subspaces $D_*^{\perp}, W_*, Z_0(v_0)$.

2. Enumerate all $v'_0 \in V'$, construct their analogous subspaces, and retain those choices with the same codimensions d, f, b.

3. Choose bases adapted to the two triples of subspaces. Enumerate every invertible map on the corresponding three quotients, of dimensions d, f, b. For each choice normalize those quotient maps, include the bilinear encoding of the prescribed subspaces, and invoke Lemma 3.6 on the target and first-input polynomials, in that order. Return a successful triple after checking it on all the input data. If no choice succeeds, reject.

To prove completeness, suppose $g = (g_X, g_V, g_Z)$ transports the two original inputs. The matrix spaces satisfy

$$A'_{gZz} = g_X^{-\mathsf{t}} A_z g_V^{-1}.$$

The vector $v'_0 = g_V v_0$ is among those enumerated, and

$$Z'_0(v'_0) = g_Z Z_0(v_0), \qquad \mathcal{K}'_{v'_0} = g_X^{-\mathsf{t}} \mathcal{K}_{v_0} g_V^{-1}.$$

Equation (3.4) transports W_* and D_*, and therefore transports $D_*^{\perp}$ by g_X. This choice has the same three codimensions. The three induced quotient maps of g occur in the enumeration. Once they are

fixed, its remaining maps have exactly the shape (3.9). Its unique extension to the spaces in (3.15) is also permitted. Lemma 3.6 therefore finds a solution on this choice. Conversely, a solution of that lemma matches the complete tensor and the listed forms, and its projection matches all prescribed subspaces by (3.15). Thus every returned triple has the required properties.

The enumeration of original vectors and the hyperplane calculations cost $q^{O(n)}\operatorname{poly}(L,\log q)$. For each retained target vector, the quotient-map enumeration has at most $q^{b^2+d^2+f^2}$ choices. The fixed-quotient algorithm costs $\operatorname{poly}(L,n,\log q,p)$. Since $p\le q$ and $n\ge 1$, its fixed polynomial dependence on p is absorbed by $q^{O(n)}$. Equation (3.6) proves the total bound $q^{O(n+a^2)}\operatorname{poly}(L,\log q)$. The construction of the original subspaces takes place only in X,V,Z; the additional spaces from the prescribed-subspace encoding do not enter these enumerations. The bound uses $a=\operatorname{ark}(T)$ for the first input. Every retained choice for the second input has the same b,d,f, so the same bound applies to every second input.

Finally, we compute the complete stabilizer using this decision and search algorithm. Choose a basis $\alpha_1,\dots,\alpha_s$ of $V_1^*\oplus V_2^*\oplus V_3^*$, with every basis covector supported on one factor; $s\le 3n$. Let H_i be the subgroup of the desired stabilizer fixing $\alpha_1,\dots,\alpha_{i-1}$ for the dual action $g\cdot\alpha=\alpha\circ g^{-1}$. For every β in the same dual factor as α_i, call the algorithm just proved with both tensors equal to T, all the original prescribed data, the additional equalities fixing the earlier covectors, and the equation

$$\beta\circ g=\alpha_i.$$

A successful call returns $g_{i,\beta}\in H_i$ with $g_{i,\beta}\cdot\alpha_i=\beta$. Let R_i be the set of these returned elements. Then

$$H_i=\langle R_i,H_{i+1}\rangle.$$

Indeed, for any $h\in H_i$, choose the returned element with $\beta=h\cdot\alpha_i$. The element $g_{i,\beta}^{-1}h$ fixes α_i and all earlier covectors, and hence belongs to H_{i+1}. After fixing the full dual basis, $H_{s+1}=1$. It follows that $\bigcup_i R_i$ generates the complete stabilizer.

There are at most $3nq^n$ calls, and the extra linear forms do not change T or $\operatorname{ark}(T)$. The total running time is therefore still $q^{O(n+a^2)}\operatorname{poly}(L,\log q)$. Only covectors of the original three factors are enumerated. The maps on every added space are determined by the original triple, so no further basis is needed. Together with a first-input-to-second-input witness, these generators describe the full transporter coset. This proves Theorem 3.1.

4 Tensor orbits under algebra unit groups

We first give an algorithm for an affine action on a tensor product. The group is specified by generators in a faithful matrix representation, and its actions on the three tensor factors may be correlated. Throughout this section logarithms have base two. An algebra is associative and contains the identity of its represented module.

4.1 Composition series and group computations

Lemma 4.1. *Let $E\le \operatorname{End}_{\mathbb{F}_q}(V)$ be given by a basis, with $\dim V\le n$. A composition series of the E-module V, together with the induced matrix representations on its composition factors, can be computed deterministically in $q^{O(n)}\operatorname{poly}(\log q)$ time. For a composition factor S, the image $E_S\le\operatorname{End}(S)$ is a simple algebra. If every simple factor of $E/J(E)$ has dimension at most D over $\mathbb{F}_q$, then $\dim E_S\le D$.*

Proof. Consider a nonzero quotient module V/V_0 occurring during the construction. For every nonzero vector v in this quotient compute Ev by applying a basis of E and taking its linear span. This span is an E-submodule, since E is an algebra. Choose a vector for which $\dim Ev$ is smallest. The module Ev is simple: a nonzero vector in a proper nonzero submodule would generate a smaller nonzero cyclic submodule. Its inverse image extends the already constructed series. Repeat on the quotient. There are at most n iterations, each enumerating at most q^n vectors and using polynomially many field operations per vector. Bases for the inverse images and quotient representations are obtained by linear algebra.

The image E_S acts faithfully and irreducibly on S. Its radical annihilates S, so it is zero. A semisimple algebra with a faithful irreducible module has only one simple factor. Over a finite field it has the form $\mathrm{M}(a, \mathbb{F}_{q^f})$, of dimension fa^2 over $\mathbb{F}_q$. Equivalently, E_S is the simple quotient of E associated with S, proving the dimension assertion. Its dimension can also be read directly from the span of the induced matrices of a basis of E. □

We use the standard deterministic membership algorithm for permutation groups [Ser03]: given generators for a subgroup of the symmetric group on a set of size s and another permutation, one can decide whether that permutation belongs to the subgroup in time polynomial in s and the input length. The permutations are given explicitly by their values on the s points.

In particular, if $G \le \mathrm{GL}(m, q)$, we may let G act on all q^m vectors of its natural module. This action is faithful. Its explicit permutation representation is constructed by applying each matrix generator to every vector. The membership algorithm therefore takes $q^{O(m)}$ time, with polynomial dependence on the number of input generators and on $\log q$. Alternatively, for a faithfully represented subgroup of a product of matrix groups, the disjoint union of the corresponding natural vector sets is a faithful permutation domain.

We will repeatedly replace a generating list by a shorter sublist. Scan the list, retaining a generator precisely when it does not belong to the subgroup generated by those already retained. This preserves the generated group. There are at most

$$\log |\mathrm{GL}(m, q)| \le m^2 \log q \tag{4.1}$$

retained generators. Thus a generating list of any subgroup encountered below can be shortened within the same $q^{O(m)}$ cost, times a polynomial in its input length. Every retained generator is stored both as an original matrix and as a straight-line program in the original input generators.

The next lemma gives the additive kernel computation needed for affine actions. It applies to any subgroup of the original matrix group.

Lemma 4.2. *Let $K \le \mathrm{GL}(m, q)$ be given by generators, let $q = p^e$, and let $\eta : K \to U$ be a homomorphism into an explicitly represented $\mathbb{F}_p$-vector space, given by its values on the supplied generators. Then a basis of $\eta(K)$, lifts of its basis vectors, and generators of $\ker \eta$ can be computed deterministically in*

$$q^{O(m)} \operatorname{poly}(\dim_{\mathbb{F}_p} U, \log q, \textit{input length})$$

time. Given a vector in $\eta(K)$, the algorithm also constructs an element having that image within this bound.

Proof. Let S be the supplied generating list of K. The image is the $\mathbb{F}_p$-span of $\{\eta(s) : s \in S\}$. Select $b_1, \dots, b_d \in S$ whose images form a basis. For each $s \in S$, write

$$\eta(s) = \sum_{i=1}^{d} c_{s,i} \eta(b_i), \qquad c_{s,i} \in \{0, \dots, p-1\}, \qquad w_s = b_1^{c_{s,1}} \cdots b_d^{c_{s,d}}.$$

The same expression constructs a lift of any vector in the image. Powers are computed by repeated squaring. The value of η on any straight-line program is obtained by addition and negation of the values on its arguments.

Let K_0 be the normal closure in K of the following elements:

$$sw_s^{-1} \quad (s \in S), \qquad [b_i, b_j] \quad (1 \le i < j \le d), \qquad b_i^p \quad (1 \le i \le d). \tag{4.2}$$

They all lie in $\ker \eta$, so $K_0 \le \ker \eta$. The quotient K/K_0 is generated by the images of the b_i, which commute and have order dividing p. Consequently $|K/K_0| \le p^d$. On the other hand, η induces a surjection from this quotient onto its d-dimensional image, which has order p^d. These orders are equal, and $K_0 = \ker \eta$.

Here is an explicit computation of this normal closure. Start with the subgroup generated by (4.2). Shorten its generating list as above. For each current generator r and each $s \in S \cup S^{-1}$, test whether $s^{-1}rs$ belongs to the current subgroup. Adjoin an element that fails the test, shorten the list, and repeat. If all the tests succeed, the subgroup is normal in K and is exactly the required normal closure. Each addition strictly enlarges the subgroup, so (4.1) bounds the total number of additions by $m^2 \log q$. The number of current generators has the same bound. These are membership tests in the original degree-m matrix representation and therefore have the stated cost.

The list (4.2) has polynomial length. One also has $p^d = |\eta(K)| \le |K| \le q^{m^2}$, so $d \le em^2$. All conjugations, products, and powers introduce only polynomially many straight-line-program instructions per operation. No expanded word is required. □

4.2 An affine action with finite linear image

We use left actions, so that the action of gh is the action of g following that of h. Suppose $G \le \mathrm{GL}(m, q)$ acts on a vector space B by

$$g \cdot v = \lambda(g)v + c(g), \qquad c(gh) = c(g) + \lambda(g)c(h), \tag{4.3}$$

where $\lambda : G \to \mathrm{GL}(B)$ is a homomorphism.

Lemma 4.3. *Suppose the action* (4.3) *is given on generators, and* $|\lambda(G)| \le L$. *For* $a, b \in B$, *one can determine whether* $g \cdot a = b$ *for some* $g \in G$, *construct such a* g *when it exists, and compute generators of* G_a, *deterministically in*

$$L^{O(1)} q^{O(m)} \,\mathrm{poly}(\dim B, \log q, \textit{input length}).$$

All output elements are represented in the original group G.

Proof. Enumerate $P = \lambda(G)$ by breadth-first multiplication by the images of the generators and their inverses. For every $\ell \in P$, retain a representative $t_\ell \in G$, taking $t_1 = 1$. This computation uses at most L image elements. Schreier's lemma gives generators for $K = \ker \lambda$: if S is a generating set for G closed under inverses, use

$$t_\ell s t_{\ell\lambda(s)}^{-1}, \qquad \ell \in P,\ s \in S.$$

Their matrices and straight-line programs are explicit. Shorten this list using the original faithful representation.

The restriction of the affine action to K is a translation action. The map

$$\eta : K \longrightarrow B, \qquad \eta(k) = c(k),$$

is an additive homomorphism. Compute the translation space $U = \eta(K)$, lifts of an $\mathbb{F}_p$-basis, and generators of $K_0 = \ker \eta$ by Lemma 4.2.

Every element whose linear image is ℓ is uniquely in the coset Kt_ℓ. For $k \in K$,

$$(kt_\ell) \cdot a = t_\ell \cdot a + \eta(k).$$

Consequently

$$G \cdot a = \bigcup_{\ell \in P} (t_\ell \cdot a + U). \tag{4.4}$$

Test these affine subspaces by $\mathbb{F}_p$-linear algebra. If $b - t_\ell \cdot a \in U$, construct k with that displacement and return kt_ℓ.

To compute G_a, consider precisely those $\ell \in P$ for which $a - t_\ell \cdot a \in U$. Choose $k_\ell \in K$ realizing this translation and put $r_\ell = k_\ell t_\ell$. Each r_ℓ fixes a. We claim that K_0, together with these r_ℓ, generates G_a. Indeed, for any $g \in G_a$, take $\ell = \lambda(g)$. The corresponding r_ℓ exists by (4.4), and gr_ℓ^{-1} belongs to K and fixes a. Its translation is zero, so it belongs to K_0. This proves the claim and the exactness of the stabilizer computation.

The image enumeration and the Schreier list have length polynomial in L and the supplied generator count. All subsequent operations have the bounds from Lemma 4.2. A matrix for the affine action of a product or inverse is obtained from

$$(\lambda(g), c(g))(\lambda(h), c(h)) = (\lambda(g)\lambda(h), c(g) + \lambda(g)c(h)),$$
$$(\lambda(g), c(g))^{-1} = (\lambda(g)^{-1}, -\lambda(g)^{-1}c(g)).$$

Thus the original lifts and their affine actions remain available throughout the computation. □

4.3 Affine actions on tensor products

Theorem 4.4. *Let $H \leq \mathrm{GL}(m, q)$ be given by generators, where $m = O(n)$. For $i \in \{1, 2, 3\}$ let B_i have dimension at most n, and let $E_i \leq \mathrm{End}(B_i)$ be an algebra given by a basis. Suppose every simple factor of each $E_i/J(E_i)$ has dimension at most D over $\mathbb{F}_q$. Let*

$$\pi_i : H \longrightarrow E_i^\times$$

be homomorphisms, and suppose H has the affine action on $\mathcal{V} = B_1 \otimes B_2 \otimes B_3$ given by

$$h \cdot v = L(h)v + c(h), \qquad L(h) = \pi_1(h) \otimes \pi_2(h) \otimes \pi_3(h), \tag{4.5}$$

where $c(gh) = c(g) + L(g)c(h)$. The factor images and the cocycle are specified on the generators and are evaluated using straight-line programs.

For $a, b \in \mathcal{V}$, one can compute H_a and the transporter

$$\{h \in H : h \cdot a = b\},$$

represented either as the empty set or by an element and generators of H_a, deterministically in

$$q^{O(n+D)} \operatorname{poly}(\textit{input length}, \log q)$$

time.

Proof. If a factor has dimension zero, $\mathcal{V} = 0$ and the conclusion is immediate. Otherwise apply Lemma 4.1 to compute

$$0 = B_{i,0} < B_{i,1} < \cdots < B_{i,\ell_i} = B_i \quad (i = 1, 2, 3).$$

Choose vector-space complements for successive terms and write $S_{i,j} = B_{i,j}/B_{i,j-1}$. In the resulting block basis each E_i acts by upper block triangular matrices. Give the tensor blocks the lexicographic order on their triples of indices. Initial segments in this order span invariant subspaces: an image of a block with indices (j_1, j_2, j_3) has indices componentwise at most that triple. We obtain an H-invariant filtration for the linear action

$$0 = F_0 < F_1 < \cdots < F_s = \mathcal{V}, \qquad s = \ell_1\ell_2\ell_3 \le n^3, \tag{4.6}$$

whose successive quotients are of the form

$$S_{1,j_1} \otimes S_{2,j_2} \otimes S_{3,j_3}.$$

By Lemma 4.1, the image algebra on each $S_{i,j}$ is simple of dimension at most D. Its unit group has order at most q^D. The linear action on any quotient of (4.6), for any subgroup of H, therefore has image of order at most q^{3D}. Its matrix representation is computed from the three quotient representations. Correlations among the π_i can only reduce this order. The dimension of the quotient is at most n^3 and does not enter this image-order bound.

We process (4.6) in reverse order. The affine action descends to $\mathcal{V}/F_j$, since $L(H)$ preserves F_j. For each j put

$$G_j = \{g \in H : g \cdot a - a \in F_j\}.$$

Thus $G_s = H$ and $G_0 = H_a$. Suppose G_j is known. In $\mathcal{V}/F_{j-1}$, the fiber over the point $a + F_j$ is the affine space $a + F_j/F_{j-1}$, preserved by G_j. With origin a, identify it with F_j/F_{j-1}. Its action is

$$g \cdot v = L_j(g)v + c_j(g), \qquad c_j(g) = (g \cdot a - a) + F_{j-1}, \tag{4.7}$$

where $L_j(g)$ is the induced linear action on F_j/F_{j-1}. The displacement is well-defined because $g \in G_j$, and the affine action law follows either from (4.5) or from the action on the fiber. Its linear image has order at most q^{3D}. Lemma 4.3, applied to the origin, computes its complete stabilizer. This is exactly G_{j-1}. Continuing computes all the groups G_j and, in particular, H_a.

We now compute a transporter, using the same groups and fiber actions. Start with $w = 1$. At the beginning of step j maintain

$$w^{-1} \cdot b - a \in F_j. \tag{4.8}$$

This holds for $j = s$. Set $y = w^{-1} \cdot b$ and let $\overline{y} = (y - a) + F_{j-1} \in F_j/F_{j-1}$. By Lemma 4.3, decide whether an element $u \in G_j$ sends zero to $\overline{y}$ under (4.7). If it does, replace w by wu. Then $u \cdot a - y \in F_{j-1}$, so $u^{-1} \cdot y - a \in F_{j-1}$ and (4.8) holds at the next step. The target normalization is the full affine inverse,

$$w^{-1} \cdot b = L(w)^{-1}(b - c(w)). \tag{4.9}$$

To verify that a failed step proves emptiness, the invariant also gives the precise transporter set for the preceding quotient:

$$\{v \in H : v \cdot a = b \pmod{F_j}\} = wG_j.$$

Indeed applying w^{-1} and using (4.8) identifies its right-hand factor with the stabilizer of $a + F_j$. Thus any full transporter would supply an element $u \in G_j$ solving the current fiber problem. If no

such u exists, the full transporter is empty. If every step succeeds, $w \cdot a = b$ and the full transporter is wH_a.

There are at most n^3 quotients. Each has a linear image of size at most q^{3D}, so Lemma 4.3 costs $q^{O(n+D)}$ per step, times a polynomial in the input size. Every intermediate subgroup is represented on the original m-dimensional space, and its generating list is shortened using (4.1). This bounds the lengths of the group inputs to successive steps. Linear algebra in $\mathcal{V}$ and its quotients has polynomial cost because $\dim \mathcal{V} \le n^3$.

Finally, all witnesses, kernel elements, and subgroup generators are stored as original matrices with shared straight-line-program instructions. The number of such instructions is bounded by the number of arithmetic and group operations just counted. Matrix multiplication and repeated squaring evaluate the programs directly; expanding them into words is unnecessary. This proves the claimed deterministic bit complexity and the output guarantees. □

5 Isomorphism modulo a tensor subspace

We next apply Theorem 4.4 after a constrained low-analytic-rank computation. For a linear map $g : V \to V'$, the symbol g^* denotes pullback of forms. Composition is written as matrix composition on column vectors; thus $(gh)^* = h^* g^*$ whenever the composition is defined.

Let T and T' be trilinear forms on three ordered pairs of spaces V_i, V_i', with corresponding dimensions equal and at most n. Let $A_i \le V_i$ and $A_i' \le V_i'$ be prescribed subspaces of matching dimensions. Prescribe also ordered bilinear forms, ordered linear forms, and ordered subspace arrangements of the types permitted by Theorem 3.1. All maps in this section are required to match these data, including A_i with A_i'. In a source stabilizer problem the source data are used on both sides. Set

$$\mathcal{Q} = A_1^{\perp} \otimes A_2^{\perp} \otimes A_3^{\perp} \le V_1^* \otimes V_2^* \otimes V_3^*,$$

where $A_i^{\perp} = \{\alpha \in V_i^* : \alpha(A_i) = 0\}$, and define $\mathcal{Q}'$ in the same way. Thus $\mathcal{Q}$ consists of the trilinear forms that factor through all three quotients V_i/A_i.

5.1 Encoding the three restrictions

Take an abstract copy U_i of A_i and its inclusion $\iota_i : U_i \to V_i$. Define

$$\begin{aligned} F_{T,A}((v_1,u_1),(v_2,u_2),(v_3,u_3)) = {} & T(\iota_1 u_1, v_2, v_3) \\ & + T(v_1, \iota_2 u_2, v_3) + T(v_1, v_2, \iota_3 u_3) \end{aligned} \tag{5.1}$$

on the three spaces $\widetilde{V}_i = V_i \oplus U_i$, each of dimension at most $2n$. Construct the corresponding target tensor using copies U_i' and inclusions ι_i'.

In each enlarged factor prescribe the two summands and the subspace

$$\Gamma_i = \{(\iota_i u, u) : u \in U_i\},$$

paired with the corresponding target subspaces. Extend the original linear and bilinear forms by zero on the U_i summands, and include each original subspace arrangement in the direct sum of the V_i summands of the enlarged factors.

Lemma 5.1. *Projection to the original factors is a bijection between the transporter of the two tensors in* (5.1) *with these prescribed data and the triples $g_i : V_i \to V_i'$ matching the original data and satisfying*

$$g^* T' - T \in \mathcal{Q}. \tag{5.2}$$

For source stabilizers this projection is a group isomorphism.

Write T_i for the restriction of T to A_i in its ith factor. Then

$$\operatorname{ark}(F_{T,A}) \le \operatorname{ark}(T_1) + \operatorname{ark}(T_2) + \operatorname{ark}(T_3). \tag{5.3}$$

In particular, if $T + \mathcal{Q}$ contains a form of analytic rank at most α, then $\operatorname{ark}(F_{T,A}) \le 3\alpha$.

Proof. A map preserving the two enlarged summands has the form $\operatorname{diag}(g_i, h_i)$. Matching Γ_i and Γ_i' is equivalent to

$$g_i \iota_i = \iota_i' h_i.$$

Hence $g_i A_i = A_i'$, and h_i is uniquely determined by g_i. Conversely every g_i taking A_i to A_i' has this unique invertible extension. The original prescribed forms and arrangements have the same transport equations under this extension.

The three summands of (5.1) occupy distinct tensor blocks, namely $U_1V_2V_3$, $V_1U_2V_3$, and $V_1V_2U_3$. They must therefore match separately. Their equality is precisely the assertion that $g^*T' - T$ vanishes whenever at least one input belongs to its corresponding A_i. This is equivalent to (5.2), by the universal property of the three quotient spaces, or by expanding in bases adapted to the A_i. The extensions are unique and respect composition, proving both the bijection and the stabilizer statement.

Restriction, or more generally linear pullback, in one tensor factor cannot increase analytic rank. To see this, average that factor last in the bias formula. Averaging over the full factor imposes that the resulting linear functional is zero, whereas averaging over a subspace only imposes that its restriction is zero. The latter event contains the former. A general pullback has the same average as restriction to its image, since all fibers have the same size. Apply this observation successively to the factors.

Each summand of (5.1) is a factorwise pullback of one T_i. Analytic-rank subadditivity [Lov19] therefore proves (5.3). If $R \in T + \mathcal{Q}$, then R and T have the same three restrictions, so all three summands are also pullbacks of R. This proves the last assertion. □

5.2 Computing the full transporter

Proposition 5.2. *Suppose q is odd, and use the prescribed data and notation above. Let $\beta =$* $\operatorname{ark}(F_{T,A})$. *Let H be the group of triples preserving the source data and satisfying $h^*T - T \in \mathcal{Q}$. Suppose explicitly based algebras*

$$E_i \le \operatorname{End}(V_i/A_i), \qquad i = 1, 2, 3,$$

contain the induced actions of H on these quotient spaces, and every simple factor of each $E_i/J(E_i)$ has dimension at most D over $\mathbb{F}_q$.

The transporter of T to T' matching all the prescribed data can be computed deterministically in

$$q^{O(n+\beta^2+D)} \operatorname{poly}(\textit{input length}, \log q) \tag{5.4}$$

time. The output is either the empty set or one isomorphism together with generators of the complete constrained source stabilizer.

Proof. Apply Theorem 3.1 to the enlarged tensors and all the data in Lemma 5.1. Its factor dimensions are at most $2n$. It either rejects or returns one transporter and generators of the complete source stabilizer. Project these maps to the original factors. By Lemma 5.1, the result is exactly a transporter g_0 satisfying (5.2) and generators of H. Every solution of (5.2) is in the coset g_0H.

The cost is $q^{O(n+\beta^2)}\,\mathrm{poly}(\text{input length}, \log q)$. Theorem 3.1 bounds this computation in terms of the source parameter β for every target input.

Represent H faithfully by its original three factor matrices, as a subgroup of $\mathrm{GL}(\dim V_1 + \dim V_2 + \dim V_3, q)$. Define the left action on trilinear forms by

$$\rho(h) = (h_1^{-1})^* \otimes (h_2^{-1})^* \otimes (h_3^{-1})^*.$$

In coefficient column coordinates its matrix is $h_1^{-\mathsf{t}} \otimes h_2^{-\mathsf{t}} \otimes h_3^{-\mathsf{t}}$. Since H preserves each A_i, it preserves $\mathcal{Q}$. Its restriction to this space depends only on the induced quotient maps $\overline{h}_i$:

$$\rho_{\mathcal{Q}}(h) = \overline{h}_1^{-\mathsf{t}} \otimes \overline{h}_2^{-\mathsf{t}} \otimes \overline{h}_3^{-\mathsf{t}}$$

under $A_i^{\perp} \simeq (V_i/A_i)^*$.

The property $h^*T - T \in \mathcal{Q}$ also holds for h^{-1}. Consequently

$$c(h) = \rho(h)T - T \in \mathcal{Q}.$$

It satisfies

$$c(hk) = c(h) + \rho_{\mathcal{Q}}(h)c(k),$$

by the representation law for ρ. Thus

$$h \cdot z = \rho_{\mathcal{Q}}(h)z + c(h) \qquad (z \in \mathcal{Q}) \tag{5.5}$$

is an affine action of the known group H.

For each i, the transpose algebra E_i^{t} on $(V_i/A_i)^*$ is isomorphic to the opposite algebra of E_i, so its simple factors have the same dimensions. The matrices $\overline{h}_i^{-\mathsf{t}}$ belong to $(E_i^{\mathsf{t}})^{\times}$: the inverse of an invertible element of a unital matrix algebra belongs to that algebra, for example by the Cayley–Hamilton theorem. The three maps $h \mapsto \overline{h}_i^{-\mathsf{t}}$ are homomorphisms. The action (5.5) therefore satisfies all the hypotheses of Theorem 4.4, with tensor factors $(V_i/A_i)^*$ and the original faithful representation of H of degree at most $3n$. Its generator images and displacements are computed by linear algebra and evaluation on T.

Set $y = g_0^*T' - T \in \mathcal{Q}$. We claim that full isomorphisms in g_0H correspond exactly to solutions of

$$h \cdot 0 = y. \tag{5.6}$$

Indeed, this equation says $\rho(h)T = g_0^*T'$. Pulling back by h gives

$$(g_0h)^*T' = h^*g_0^*T' = T.$$

Conversely this last equality, pulled back by h^{-1}, gives (5.6). All prescribed data remain matched since the correcting element belongs to H.

Apply Theorem 4.4 to zero and y. If its transporter is empty, no full isomorphism exists. Otherwise let h_0 be its witness and let K be the stabilizer of zero under (5.5). This stabilizer is exactly the complete constrained source stabilizer of T, because

$$k \in K \quad \Longleftrightarrow \quad c(k) = 0 \quad \Longleftrightarrow \quad k^*T = T.$$

The required full transporter is $(g_0h_0)K$. The affine computation costs $q^{O(n+D)}$ times polynomial input length. Together with the first computation this proves (5.4). □

The parameter in Proposition 5.2 can thus be bounded using the three restrictions separately. In particular, if their analytic ranks are at most α, Lemma 5.1 gives the bound $q^{O(n+\alpha^2+D)}$. The algorithm uses the explicit tensor $F_{T,A}$ defined in (5.1).

6 Subalgebras and restrictions of matrix spaces

This section proves the algebraic statement used to select contractions of the input tensor. Throughout the section, E is a unital $\mathbb{F}_q$-subalgebra of $\mathrm{End}_{\mathbb{F}_q}(W)$, where $\dim W = N \geq 1$. Subalgebras contain the same identity as E. All logarithms without a subscript have base two.

6.1 Finite-dimensional algebras and their modules

An E-submodule of W is a subspace invariant under every matrix in E. A nonzero module is *simple* if its only invariant subspaces are zero and the whole module. A composition series is a chain of invariant subspaces whose successive quotients are simple. The multiset of isomorphism types of these quotients is independent of the series.

The Jacobson radical $J(E)$ is the largest nilpotent two-sided ideal of E. The standard structure theorems for finite-dimensional algebras over finite fields give

$$E/J(E) \cong \prod_{i=1}^{k} A_i, \qquad A_i = \mathrm{M}(a_i, \mathbb{F}_{q^{f_i}}). \tag{6.1}$$

Moreover, E has a subalgebra L such that $E = L \oplus J(E)$ as vector spaces and the quotient map restricts to an isomorphism $L \to E/J(E)$. Such an L is called a semisimple complement. Every module for L is a direct sum of simple modules. These facts are the Wedderburn–Artin and Wedderburn–Malcev theorems; see [Pie82].

For A_i in (6.1), put

$$D_i = f_i a_i^2, \qquad \delta_i = f_i a_i. \tag{6.2}$$

Thus D_i is the dimension of the algebra A_i, whereas δ_i is the dimension of its simple module. Let d_i be the multiplicity of that simple module in a composition series of W. The representation $E \subseteq \mathrm{End}(W)$ is faithful, so every factor in (6.1) occurs, and

$$d_i \geq 1, \qquad \sum_i d_i \delta_i = N, \qquad \delta_i \geq \sqrt{D_i}. \tag{6.3}$$

For example, faithfulness implies the first assertion because an idempotent lifting the identity of an absent factor would act as zero on every composition quotient. It would then be both nilpotent and idempotent, and hence zero.

We will use the following form of the conjugacy statement for semisimple subalgebras. Its proof also specifies precisely which conjugating matrices are needed.

Lemma 6.1. *Fix a semisimple complement L of E. If $B \subseteq E$ is a semisimple subalgebra, possibly with an identity different from that of E, then there is a matrix $u \in 1 + J(E)$ such that $u^{-1}Bu \subseteq L$. Under the quotient map $E \to E/J(E)$, the subalgebras B and $u^{-1}Bu$ have the same image.*

Proof. Write $\pi : E \to E/J(E)$ for the quotient map. The ideal $B \cap J(E)$ of B is nilpotent, and therefore zero. Consequently

$$\beta = (\pi|_L)^{-1} \circ \pi|_B : B \longrightarrow L$$

is an injective algebra homomorphism. Let e and $e_0 = \beta(e)$ be the identities of B and $\beta(B)$, respectively. On the unital semisimple algebra $C = B \times \mathbb{F}_q$, define unital homomorphisms

$$\sigma(b, c) = b + c(1 - e), \qquad \eta(b, c) = \beta(b) + c(1 - e_0).$$

They have the same composition with π.

A finite-dimensional semisimple algebra over a finite field is separable. Equivalently, there are $x_j, y_j \in C$ such that

$$\sum_j x_j y_j = 1, \qquad \sum_j c x_j \otimes y_j = \sum_j x_j \otimes y_j c \quad (c \in C). \tag{6.4}$$

For a finite-field extension $K/\mathbb{F}_q$, the algebra $K \otimes_{\mathbb{F}_q} K$ is a product of copies of K. The idempotent for the component defined by multiplication $K \otimes_{\mathbb{F}_q} K \to K$ satisfies (6.4). This property is preserved by matrix algebras and finite products. For a matrix algebra over its center, the required element is $\sum_i E_{i1} \otimes E_{1i}$; combining it with a separability element for the center extension gives (6.4) over $\mathbb{F}_q$.

Set $u = \sum_j \sigma(x_j)\eta(y_j)$. Since $\pi\sigma = \pi\eta$, the first identity in (6.4) gives $u \equiv 1 \pmod{J(E)}$. Thus u is invertible. The second identity gives $\sigma(c)u = u\eta(c)$ for every $c \in C$. In particular, $u^{-1}bu = \beta(b)$ for $b \in B$, as required. □

The following procedure computes the algebra data used by the algorithm from the represented action on W.

Lemma 6.2. *Given a linear basis of $E \subseteq \mathrm{End}(W)$, one can deterministically compute a composition series of W, the induced representations on its simple quotients, the integers D_i, δ_i, d_i, and $J(E)$. One can also compute an element of E with prescribed images belonging to the distinct simple quotient algebras. The running time is*

$$q^{O(N)} \operatorname{poly}(N, \log q).$$

Proof. On any nonzero module quotient V, enumerate its vectors. For $v \neq 0$, the span Ev is obtained by applying a basis of E to v. Choose a vector for which $\dim Ev$ is positive and minimal. Then Ev is simple: a proper nonzero submodule would contain a nonzero vector generating a smaller cyclic submodule. Record this submodule, pass to the quotient, and repeat. There are at most N iterations, each using at most q^N vectors and polynomial linear algebra.

Let $\rho_j : E \to \mathrm{End}(S_j)$ be the resulting representations on simple quotients. The image $\rho_j(E)$ is a simple algebra. Indeed, its radical annihilates its simple module and hence is zero, and a semisimple algebra acting faithfully on a simple module has only one simple factor. Its $\mathbb{F}_q$-dimension D_j is computed by linear algebra, as is $\dim S_j = \delta_j$. Its center degree and matrix size follow from $f_j = \delta_j^2/D_j$ and $a_j = D_j/\delta_j$.

The kernels $K_j = \ker \rho_j$ are maximal two-sided ideals. Distinct such kernels are comaximal. Their intersection J_0 strictly lowers the composition series, so $J_0^N = 0$. The Chinese remainder theorem gives

$$E/J_0 \cong \prod_{j \in R} \rho_j(E),$$

where R contains one representative for each distinct kernel. Therefore $J_0 = J(E)$. Equal kernels identify repeated occurrences of the same simple factor and determine the multiplicities d_i. Finally, prescribing quotient images is a consistent system of linear equations in the coefficients of an element of E, by this same Chinese remainder isomorphism. □

6.2 Ideals defined by the dimensions of simple factors

Fix a positive integer τ, and let

$$\mathcal{L}_\tau(E) = \{i : D_i > \tau^2\}.$$

In $E/J(E)$, take the central idempotent which is the identity on these factors and zero on all others. For any idempotent lift $e \in E$, define

$$I_\tau(E) = EeE, \qquad W_\tau(E) = I_\tau(E)W. \tag{6.5}$$

The notation $I_\tau(E)W$ denotes the span of aw for $a \in I_\tau(E)$ and $w \in W$. It is an E-submodule.

Lemma 6.3. *The ideal and submodule in* (6.5) *are independent of the idempotent lift. For every unital subalgebra $E' \subseteq E$,*

$$I_\tau(E') \subseteq I_\tau(E), \qquad W_\tau(E') \subseteq W_\tau(E). \tag{6.6}$$

The quotient $E/I_\tau(E)$ has only simple factors of dimension at most τ^2. The ideal $I_\tau(E)$ and its submodule $W_\tau(E)$ can be computed in $q^{O(N)}\,\mathrm{poly}(N, \log q)$ time.

Proof. If e, e' are two idempotent lifts of the same quotient idempotent, put

$$c = e'e + (1 - e')(1 - e).$$

Then $c \equiv 1 \pmod{J(E)}$, so c is a unit, and $ce = e'c$. Thus $e' = cec^{-1}$, which gives $Ee'E = EeE$. The simple factors of $E/I_\tau(E)$ are precisely the factors in (6.1) not indexed by $\mathcal{L}_\tau(E)$.

To prove (6.6), choose a semisimple complement of E'. Consider one of its simple factors B of dimension greater than τ^2. The homomorphism $B \to E/I_\tau(E)$ is either zero or injective. An injective image is impossible. Indeed, this image has zero intersection with the radical of $E/I_\tau(E)$, since that intersection would be a nilpotent ideal of a simple algebra. It therefore embeds in the semisimple quotient. At least one projection to a simple factor of that quotient is injective, contradicting its dimension bound. It follows that $B \subseteq I_\tau(E)$. The sum of the identities of all these factors lies in $I_\tau(E)$, as does the two-sided E'-ideal that it generates. This proves both inclusions.

For computation, use Lemma 6.2 to find $x \in E$ whose simple quotient images are the prescribed zero or identity matrices. Then $x^2 - x \in J(E)$ and $(x^2 - x)^N = 0$. Polynomial Chinese remaindering gives a polynomial $h(t)$ with

$$h(t) \equiv 0 \pmod{t^N}, \qquad h(t) \equiv 1 \pmod{(t-1)^N}.$$

The matrix $e = h(x)$ is an idempotent with the required quotient image. The span of the products aeb, for basis elements a, b of E, is $I_\tau(E)$, and its images on a basis of W span $W_\tau(E)$. These last steps use polynomial linear algebra. □

6.3 A potential for subalgebra inclusions

Define

$$\Phi(E, W) = \sum_{i=1}^{k} d_i \delta_i \log D_i, \qquad \Psi_\tau(E, W) = \Phi(E, W) + \tau K_\tau(E), \qquad K_\tau(E) = |\mathcal{L}_\tau(E)|. \tag{6.7}$$

When the module W is fixed, we also write $\Phi(E)$ and $\Psi_\tau(E)$; the same convention applies to $P_\tau(E, W)$ defined below. These quantities depend on the represented algebra and its module, rather than a choice of composition series. Since $D_i \le N^2$ and every factor counted by $K_\tau(E)$ uses more than τ dimensions of W,

$$0 \le \Psi_\tau(E, W) \le 2N \log N + N. \tag{6.8}$$

Proposition 6.4. *For a unital subalgebra $E' \subseteq E$, both $\Phi(E, W)$ and $\Psi_\tau(E, W)$ are at least their counterparts for E'.*

More precisely, choose a semisimple complement L' of E', and form a bipartite graph. Its left vertices are the simple factors of $E/J(E)$ of dimension greater than τ^2, and its right vertices are the simple factors of L' of dimension greater than τ^2. Join two vertices when projection of the latter factor to the former is nonzero. Every right vertex has a neighbor. Put

$$s = \sum_{i \text{ left}} (\deg(i) - 1)_+, \qquad t = \sum_{j \text{ right}} (\deg(j) - 1), \qquad z = |\{i \text{ left} : \deg(i) = 0\}|.$$

Then

$$\Phi(E, W) - \Phi(E', W) \geq 2\tau s, \tag{6.9}$$

$$\Psi_\tau(E, W) - \Psi_\tau(E', W) \geq \tau(s + t + z). \tag{6.10}$$

Proof. The intersection $L' \cap J(E)$ is a nilpotent ideal of the semisimple algebra L', and hence is zero. Thus L' embeds in $E/J(E)$. Each simple factor B of L' has a nonzero, and therefore injective, projection to some old simple factor. If $\dim B > \tau^2$, that old factor also has dimension greater than τ^2. This proves the assertion about right vertices.

Consider an old factor $A_i = \mathrm{M}(a_i, \mathbb{F}_{q^{f_i}})$ and one copy of its simple module. For every simple factor B of L' acting nontrivially on this module, let e_B be the image of its identity in A_i and let t_B be the rank of e_B over $\mathbb{F}_{q^{f_i}}$. These idempotents are orthogonal and sum to the identity. Therefore

$$\sum_B t_B = a_i, \qquad \dim_{\mathbb{F}_q} e_B S = f_i t_B, \qquad \dim_{\mathbb{F}_q} B \leq f_i t_B^2. \tag{6.11}$$

The last inequality holds because B injects into the corner $e_B A_i e_B$.

Restrict an E-composition series of W to L' and decompose its quotients as L'-modules. This gives the same multiset of simple L'-modules as restricting an E'-composition series of W. The contribution of the chosen old simple module to the difference of the two Φ values is consequently

$$\sum_B f_i t_B \log \frac{f_i a_i^2}{\dim B}. \tag{6.12}$$

Every term is nonnegative, proving monotonicity of Φ.

Suppose the left vertex A_i has $h \geq 2$ neighbors. Apart from a neighbor with largest t_B, all other neighbors have $t_B \leq a_i/2$. For each of these $h - 1$ factors, (6.11) and $\dim B > \tau^2$ imply $f_i t_B > \tau$, and its term in (6.12) is at least

$$f_i t_B \log \frac{a_i^2}{t_B^2} \geq 2 f_i t_B > 2\tau.$$

At least one old simple-module copy occurs for each left vertex, by faithfulness. Summing these inequalities proves (6.9).

Let k and k' be the numbers of left and right vertices. Counting edges on the two sides gives $k' - k = s - t - z$. Therefore

$$\Psi_\tau(E, W) - \Psi_\tau(E', W) = \Phi(E, W) - \Phi(E', W) + \tau(k - k') \geq \tau(s + t + z),$$

which proves (6.10) and monotonicity of Ψ_τ. □

The potential is compared exactly using integers. Define

$$P_\tau(E,W) = 2^{\tau K_\tau(E)} \prod_i D_i^{d_i \delta_i}. \tag{6.13}$$

It has $O(N \log(N+2))$ bits and $\log P_\tau(E,W) = \Psi_\tau(E,W)$. In particular, for an integer r, the inequality $\Psi_\tau(E,W) - \Psi_\tau(E',W) \geq r$ is equivalent to $P_\tau(E,W) \geq 2^r P_\tau(E',W)$.

6.4 Large simple subalgebras of matrix algebras

Lemma 6.5. *Suppose $B = \mathrm{M}(m, \mathbb{F}_{q^g})$ embeds as an $\mathbb{F}_q$-algebra into $A = \mathrm{M}(a, \mathbb{F}_{q^f})$. Its identity may be an idempotent of A. Write $h = \gcd(f,g)$. Then*

$$a \geq \frac{mg}{h}, \qquad \frac{\dim_{\mathbb{F}_q} B}{\dim_{\mathbb{F}_q} A} \leq \frac{h^2}{fg}. \tag{6.14}$$

If $\dim_{\mathbb{F}_q} B > \frac{1}{2} \dim_{\mathbb{F}_q} A$, then $g = f$ and $m > a/\sqrt{2}$. In this case, if e is the identity of B, the subspace $e\mathbb{F}_{q^f}^a$ has dimension exactly m over $\mathbb{F}_{q^f}$, and the image of B on this subspace is its entire $\mathbb{F}_{q^f}$-endomorphism algebra.

Proof. Put $F = \mathbb{F}_{q^f}$ and let $V = eF^a$. On V, the action of B commutes with scalar multiplication by F. Hence V is a module for

$$F \otimes_{\mathbb{F}_q} \mathrm{M}(m, \mathbb{F}_{q^g}) \cong \prod_{j=1}^{h} \mathrm{M}(m, \mathbb{F}_{q^{\mathrm{lcm}(f,g)}}).$$

Indeed, an irreducible degree-g polynomial over $\mathbb{F}_q$ splits over F into h irreducible factors of degree g/h: its roots are permuted in cycles of that length by the q^f-power Frobenius map. This gives the displayed field product. Each simple module of this algebra has dimension mg/h over F. As $V \neq 0$, this proves the first inequality in (6.14). The second follows by substituting it into $\dim B = gm^2$ and $\dim A = fa^2$.

Write $f = hu$ and $g = hv$, with $\gcd(u,v) = 1$. If $f \neq g$, then $uv \geq 2$, so $h^2/(fg) \leq 1/2$. The strict dimension hypothesis therefore forces $f = g$, and then $m > a/\sqrt{2} > a/2$. The dimension of V over F is a positive multiple of m and is at most a, so it equals m. The image of B lies in $\mathrm{End}_F(V)$ and both algebras have $\mathbb{F}_q$-dimension fm^2. They are equal. □

In the last case of the lemma, an $\mathbb{F}_q$-linear map on V commuting with B is multiplication by an element of $\mathbb{F}_{q^f}$. Indeed, it first commutes with all F-scalar maps and so is F-linear, and then commuting with the matrix units makes it scalar. On d identical copies of V, the commuting maps are $d \times d$ matrices of such scalar maps.

6.5 Analytic rank on the subspace generated by an ideal

Let $\mathcal{S} \subseteq \mathrm{End}(W)$ be a linear matrix space, and let $A \subseteq W$ be a subspace. We associate to the restricted matrices the trilinear form

$$(M, a, \lambda) \longmapsto \lambda(Ma), \qquad M \in \mathcal{S}, \quad a \in A, \quad \lambda \in W^*.$$

We denote its analytic rank by $\mathrm{ark}(\mathcal{S}|_A)$. The usual rank formula for tensor bias gives

$$q^{-\mathrm{ark}(\mathcal{S}|_A)} = \mathop{\mathbb{E}}_{M \in \mathcal{S}} q^{-\mathrm{rank}(M|_A)}. \tag{6.15}$$

A linear parametrization of $\mathcal{S}$ by a larger coefficient space gives the same average, because all fibers have the same cardinality.

Theorem 6.6. *Let $r \geq 1$ be an integer and set $\tau = 10r$. For $M \in \mathcal{S}$, write*

$$E_M = E \cap \{a \in \mathrm{End}(W) : aM = Ma\}.$$

At least one of the following statements holds:

1. *There is an $M \in \mathcal{S}$ such that*

$$\Psi_\tau(E, W) - \Psi_\tau(E_M, W) \geq r.$$

2. *There is an $M \in \mathcal{S}$ such that*

$$\dim W_\tau(E) - \dim W_\tau(E_M) \geq r.$$

3. *The restricted matrix space satisfies*

$$\mathrm{ark}(\mathcal{S}|_{W_\tau(E)}) \leq r + \sum_{i \in \mathcal{L}_\tau(E)} f_i d_i^2 \leq r + \frac{N^2}{100r^2}. \tag{6.16}$$

More precisely, if the first two statements fail, there is a linear subspace $\mathcal{S}_0 \subseteq \mathcal{S}$ of codimension at most $\sum_{i \in \mathcal{L}_\tau(E)} f_i d_i^2$ such that every $M \in \mathcal{S}_0$ satisfies $\mathrm{rank}(M|_{W_\tau(E)}) < r$.

Proof. Suppose the first two statements fail. Put

$$A = W_\tau(E), \qquad A_M = W_\tau(E_M), \qquad \ell_M = \dim A - \dim A_M < r.$$

The inclusion $A_M \subseteq A$ follows from Lemma 6.3. If $\mathcal{L}_\tau(E)$ is empty, then $A = 0$ and the conclusion follows immediately. We assume it is nonempty.

The simple factors of E_M. Fix M temporarily and apply Proposition 6.4 to $E_M \subseteq E$. Its integers s, t, z satisfy

$$\tau(s + t + z) \leq \Psi_\tau(E, W) - \Psi_\tau(E_M, W) < r < \tau.$$

Thus $s = t = z = 0$. The graph in that proposition is a matching containing all its vertices. In particular, each $i \in \mathcal{L}_\tau(E)$ has exactly one corresponding simple factor B_i of E_M of dimension greater than τ^2, and these are all the simple factors of E_M exceeding that dimension. Moreover $K_\tau(E_M) = K_\tau(E)$, so

$$\Delta\Phi_M := \Phi(E, W) - \Phi(E_M, W) < r. \tag{6.17}$$

The dimension of B_i is greater than $D_i/2$. Otherwise every new simple factor acting on an old A_i-module would have dimension at most $D_i/2$: B_i by this supposition, and all the others because their dimensions are at most $\tau^2 < \dim B_i$. Formula (6.12) would then charge at least $\delta_i \log 2 = \delta_i > \tau$ to just one old simple-module copy, contradicting (6.17).

Lemma 6.5 shows that

$$B_i \cong \mathrm{M}(m_i, \mathbb{F}_{q^{f_i}}), \qquad m_i > a_i/\sqrt{2},$$

and its identity has rank exactly m_i over $\mathbb{F}_{q^{f_i}}$ on an old natural module. The factor B_i has zero projection to every other old factor: projection to another factor of dimension greater than τ^2 is excluded by the matching, and injection into a factor of dimension at most τ^2 is excluded by dimension.

Every new simple factor acting on an old A_i-module has dimension at most $\dim B_i = f_i m_i^2$. The contribution of all d_i old copies to $\Delta\Phi_M$ is therefore at least

$$f_i a_i d_i \log \frac{f_i a_i^2}{f_i m_i^2} = 2 f_i a_i d_i \log \frac{a_i}{m_i} \geq 2 f_i d_i (a_i - m_i).$$

Here $\log(a_i/m_i) \geq (a_i - m_i)/a_i$. Summing and using nonnegativity of the other contributions gives

$$k_M := \sum_{i \in \mathcal{L}_\tau(E)} f_i d_i (a_i - m_i) \leq \frac{\Delta\Phi_M}{2} < \frac{r}{2}. \tag{6.18}$$

A fixed composition series. Fix a semisimple complement L of E. In (6.5), choose $e \in L$. Choose an E-composition series

$$0 = W_0 \subset W_1 \subset \cdots \subset W_v = W$$

which contains A as one of its terms. Choose L-invariant complements to the successive subspaces. Such complements exist because L is semisimple. Use identical module coordinates on copies of the same old simple type. In the resulting block basis, L is block diagonal, E is upper block triangular, and every element of $1 + J(E)$ is upper block unitriangular. The subspace A is an initial segment of these blocks.

The matrix e is the identity on blocks whose types are indexed by $\mathcal{L}_\tau(E)$ and zero on all other blocks. Every block of an indexed type belongs to A: the quotient W/A is annihilated by $I_\tau(E)$ and has only the other simple types.

Choose a semisimple complement of E_M and take the product of its factors B_i just identified. Lemma 6.1 supplies $u = u_M \in 1 + J(E)$ which moves this product into L. Write

$$E' = u^{-1} E_M u, \qquad M' = u^{-1} M u, \qquad A' = u^{-1} A_M,$$

and denote the resulting subalgebra of L by B. Let e_B be its identity. The matrices u preserve A, and

$$A' = E' e_B W \subseteq A, \qquad e_B W \subseteq A', \qquad e e_B = e_B e = e_B, \qquad \dim A - \dim A' = \ell_M. \tag{6.19}$$

The choices of u, B, and A' may depend on M.

Maps on the copies of the simple modules. The map M' commutes with B. On $e_B W$, its restriction is consequently a $d_i \times d_i$ matrix over $\mathbb{F}_{q^{f_i}}$ for each $i \in \mathcal{L}_\tau(E)$, acting on the d_i identical copies of the m_i-dimensional natural module. Extend each of these scalar blocks from m_i to a_i natural coordinates. This defines $w \in \mathrm{End}(W)$ commuting with L, zero on all other old simple types, and satisfying

$$M' e_B = w e_B, \qquad w e_B = e_B w. \tag{6.20}$$

These matrices w belong to the space of L-endomorphisms supported on the indexed types, whose dimension is

$$C = \sum_{i \in \mathcal{L}_\tau(E)} f_i d_i^2. \tag{6.21}$$

Their blocks between copies of the same indexed type are multiplication by elements of its center field. All their other blocks are zero.

Adjoin the complementary scalar identity to B. This gives a unital semisimple algebra acting on A and its submodule A'. Choose a module projection $p : A \to A'$ with $p|_{A'} = 1$, and put $G = M'p$. The identities $A' = E'e_BW$ and $M'e_B = e_BM'$ give $M'A' = E'e_BM'W \subseteq A'$. The projection has kernel of dimension ℓ_M, so

$$\operatorname{rank}(M'|_A - G) \le \ell_M, \qquad \operatorname{im} G = M'A' = E'M'e_BW = E'we_BW. \tag{6.22}$$

The equalities of images use surjectivity of p and (6.19).

Replace the output components of G on eW by those of w, by setting

$$H = (1-e)G + w|_A, \qquad R_1 = G - H = eG - w|_A.$$

Equivariance of p gives $e_Bp = pe_B = e_B$ on A, because p fixes e_BW. Using (6.20), we obtain

$$e_BG = M'e_B = we_B = e_Bw.$$

Thus $e_BR_1 = 0$, while $\operatorname{im} R_1 \subseteq eW$. Its image is contained in $(e - e_B)W$, whose dimension is k_M. There is consequently a single map $R : A \to W$ such that

$$M'|_A = H + R, \qquad \operatorname{rank} R \le \ell_M + k_M < 2r. \tag{6.23}$$

Both G and H have image in A. Returning to the fixed coordinates gives

$$M|_A = uH(u|_A)^{-1} + uR(u|_A)^{-1}. \tag{6.24}$$

Controlling the output rows. Suppose the output block rows of w after position j are all zero. Then $\operatorname{im} w \subseteq W_j$. Since every element of E' preserves W_j, the image identity in (6.22) gives $\operatorname{im} G \subseteq W_j$. The matrix e preserves the same flag, and therefore

$$\operatorname{im} H \subseteq W_j. \tag{6.25}$$

It suffices to know the stated vanishing for the rows of indexed simple types, since all other rows of w are zero by definition. Also

$$eH = w|_A. \tag{6.26}$$

Hence a row of H of an indexed type has nonzero blocks only in columns of that same type.

Linear equations on the matrix space. We now construct nested linear subspaces of $\mathcal{S}$. Process the output block rows of indexed types in decreasing order. In each such row, process the input columns of the same type in increasing order. At each step maintain the following property: for every M in the current linear subspace, every choice of the normalization and projection above has all previously processed coefficients of w equal to zero.

Let (i, j) be the current position, with both blocks of old type A_t. All later indexed rows of w are zero, so (6.25) says that $H_{k,l} = 0$ for all $k > i$. In row i, the blocks with $l < j$ are zero by previous steps or by (6.26). Upper block unitriangularity now gives

$$\big(uH(u|_A)^{-1}\big)_{i,j} = \sum_{k \ge i,\, l \le j} u_{i,k} H_{k,l} \big((u|_A)^{-1}\big)_{l,j} = w_{i,j}. \tag{6.27}$$

Both the input and output blocks occur in A, since their types are indexed. Their common dimension over $\mathbb{F}_q$ is $\delta_t = f_t a_t$.

Let $\mu_t(\lambda)$ be multiplication by $\lambda \in \mathbb{F}_{q^{f_t}}$ on the old natural module. Equations (6.24) and (6.27) show that the actual block $M_{i,j}$ satisfies

$$\operatorname{rank}\big(M_{i,j} - \mu_t(\lambda)\big) < 2r \tag{6.28}$$

for the field coefficient λ of $w_{i,j}$. A nonzero difference of two field multiplication maps has rank $\delta_t > \tau = 10r$. Thus the coefficient in (6.28) is unique: two choices would bound that rank by less than $4r$. It is linear in M on the current subspace. Indeed, comparing (6.28) for $M_1, M_2, M_1 + M_2$ bounds the rank of any failure of additivity by less than $6r$, and scalar homogeneity follows in the same way. The coefficient is therefore an intrinsic $\mathbb{F}_q$-linear map into $\mathbb{F}_{q^{f_t}}$.

Pass to its kernel, of codimension at most f_t. Uniqueness ensures that the coefficient is zero for every allowed normalization, preserving the stated property for the next step. There are d_t^2 positions of type A_t, so the total codimension is at most C from (6.21).

On the resulting subspace $\mathcal{S}_0$, all coefficients of w vanish. By (6.22), $M'A' = 0$. Equivalently, $MA_M = 0$. Since A_M has codimension $\ell_M < r$ in A, this proves

$$\operatorname{rank}(M|_A) < r \qquad (M \in \mathcal{S}_0).$$

The subspace $\mathcal{S}_0$ has density at least q^{-C} in $\mathcal{S}$. Using (6.15) gives $\operatorname{ark}(\mathcal{S}|_A) \le C + r$.

Finally, set $N_i = f_i a_i d_i$. Since $D_i = f_i a_i^2 > 100r^2$ on the indexed types,

$$C = \sum_{i \in \mathcal{L}_\tau(E)} \frac{N_i^2}{f_i a_i^2} \le \frac{(\sum_i N_i)^2}{100r^2} \le \frac{N^2}{100r^2}.$$

This completes the proof. □

Corollary 6.7. *Suppose a linear map $x \mapsto M_x$ from $\mathbb{F}_q^s$ to $\operatorname{End}(W)$ is given by its coefficient matrices. In $q^{O(N+s)}\operatorname{poly}(N, \log q)$ time, one can find an x satisfying either of the first two inequalities in Theorem 6.6, or determine that its third conclusion holds for the entire parametrized matrix space.*

Proof. Enumerate $x \in \mathbb{F}_q^s$. Compute E_{M_x} by linear equations and use Lemmas 6.2 and 6.3 to compute its composition data and the dimension of $W_\tau(E_{M_x})$. Compare the integer potentials in (6.13) and the two subspace dimensions. If none of these tests succeeds, Theorem 6.6 applies. The algorithm uses the quotient representations and the ideal subspaces computed by those lemmas; the complements and projections in the theorem's proof establish the analytic-rank bound. □

7 The algorithm for arbitrary tensors

We now prove Theorem 1.1. Let T and T' be trilinear forms on $V_1 \times V_2 \times V_3$ and $V_1' \times V_2' \times V_3'$, respectively, with $\dim V_i = \dim V_i'$. If a factor has dimension zero, both tensors are zero and any three invertible maps give an isomorphism. Henceforth the factor dimensions are positive. Put

$$U = V_1 \oplus V_2 \oplus V_3, \qquad W = U \oplus U^*, \qquad \ell = \dim U, \qquad N = 2\ell.$$

Choose an integer $r \ge 1$ and put $\tau = 10r$. This value of τ and this representation space W will be used throughout the algorithm.

7.1 Contractions and their adjoint algebra

For $h = (x_1, x_2, x_3) \in U$, define the alternating form Q_h on U by

$$\begin{aligned} Q_h(v, w) = {} & T(v_1, w_2, x_3) - T(w_1, v_2, x_3) \\ & + T(v_1, x_2, w_3) - T(w_1, x_2, v_3) \\ & + T(x_1, v_2, w_3) - T(x_1, w_2, v_3). \end{aligned}$$

The map $h \mapsto Q_h$ is linear. Its three upper off-diagonal blocks are the bilinear forms $T(-, -, x_3)$, $T(-, x_2, -)$, and $T(x_1, -, -)$, in that order. Matching Q_h by a block diagonal map on U is equivalent to matching these three ordered bilinear forms.

Given selected vectors $h_1, \ldots, h_t$, let

$$E = \{(a, b) : a, b \in \textstyle\prod_{i=1}^{3} \operatorname{End}(V_i), \quad a^{\mathsf{t}} Q_{h_j} = Q_{h_j} b \text{ for } j \in [t]\}. \tag{7.1}$$

The multiplication is $(a, b)(a', b') = (aa', b'b)$. The map

$$(a, b) \longmapsto \operatorname{diag}(a, b^{\mathsf{t}}) \in \operatorname{End}(W)$$

is a faithful algebra representation. We identify E with its image in this representation. A basis of E is found by solving the linear equations in (7.1).

For $h \in U$, let

$$M_h = \begin{pmatrix} 0 & 0 \\ Q_h & 0 \end{pmatrix} \in \operatorname{End}(W).$$

Direct multiplication shows that $\operatorname{diag}(a, b^{\mathsf{t}})$ commutes with M_h if and only if $b^{\mathsf{t}} Q_h = Q_h a$. Transposing and using $Q_h^{\mathsf{t}} = -Q_h$ makes this equivalent to $a^{\mathsf{t}} Q_h = Q_h b$. Thus selecting h replaces E by precisely

$$E_h = E \cap \operatorname{Cent}(M_h). \tag{7.2}$$

7.2 Selecting the source vectors

Use the ideal $I_\tau(E)$, its image $W_\tau(E)$, and the potential $\Psi_\tau(E)$ defined in Section 6. Initially the list of vectors is empty, and E is the algebra (7.1) with no equations. At each step enumerate $h \in U$ in a fixed coordinate order. Select the first h for which

$$\Psi_\tau(E) - \Psi_\tau(E_h) \geq r \quad \text{or} \quad \dim W_\tau(E) - \dim W_\tau(E_h) \geq r. \tag{7.3}$$

Append it to the list, replace E by E_h, and start a new enumeration. Stop when no vector satisfies either inequality.

The comparisons in (7.3) are exact integer computations. For a composition series of W as an E-module, let D_S be the dimension of the simple algebra associated with a composition factor S, and let $K_\tau(E)$ be the number of distinct simple factors of $E/J(E)$ of dimension greater than τ^2. Then

$$P_\tau(E) = 2^{\tau K_\tau(E)} \prod_S D_S^{\dim S} \quad \text{satisfies} \quad \log P_\tau(E) = \Psi_\tau(E).$$

Accordingly the first comparison is $P_\tau(E) \geq 2^r P_\tau(E_h)$. These integers have $O(N \log(N + 2))$ bits, since $\tau K_\tau(E) < N$ and $D_S \leq N^2$.

The monotonicity proved in Section 6 applies at every inclusion $E_h \leq E$. Both terms of

$$\Theta(E) = \Psi_\tau(E) + \dim W_\tau(E)$$

are nonincreasing, and a selected vector decreases their sum by at least r. Since $0 \le \Theta(E) \le 2N \log N + 2N$, the final number of vectors satisfies

$$t \le \frac{2N \log N + 2N}{r}. \tag{7.4}$$

Each enumeration has q^{ℓ} vectors. Computing an algebra in (7.2), its composition data, and its ideal image takes $q^{O(N)}\,\mathrm{poly}(\log q)$ time by the algebra computations in Section 6. Thus all source vectors and the final algebra are found in $q^{O(N)}\,\mathrm{poly}(\log q)$ time.

At termination, the relative analytic rank theorem of Section 6, applied to the linear map $h \mapsto M_h$, gives

$$\mathrm{ark}\big((h,w) \mapsto M_h w :\ U \times W_\tau(E) \to W\big) \le \alpha, \qquad \alpha = r + \frac{N^2}{100 r^2}. \tag{7.5}$$

The coefficient map may have a kernel; averaging over it only repeats every operator equally often, so it does not change the bound.

7.3 The subspaces in the three factors

The six spaces V_i and V_i^* are invariant under E. The image of its ideal is consequently the direct sum of its images on these spaces. Define

$$A_i = I_\tau(E) V_i \le V_i \qquad (i \in [3]). \tag{7.6}$$

Let R_i be T with its ith argument restricted to A_i. We claim that

$$\mathrm{ark}(R_i) \le \alpha \qquad (i \in [3]). \tag{7.7}$$

For example, the block of $M_h|_{W_\tau(E)}$ from A_1 to V_3^* is, up to sign, the matrix of $T(-, x_2, -)|_{A_1 \times V_3}$. Therefore, for every x_1, x_2, x_3,

$$\mathrm{rank}\big(M_{(x_1,x_2,x_3)}|_{W_\tau(E)}\big) \ge \mathrm{rank}\big(T(-,x_2,-)|_{A_1 \times V_3}\big).$$

Average q to the negative of these ranks over all three x_i. On the right only x_2 matters. Formula (2.1) gives $\mathrm{bias}((h,w) \mapsto M_h w) \le \mathrm{bias}(R_1)$, proving the assertion for $i = 1$. The blocks from A_2 to V_3^* and from A_3 to V_2^* prove it for $i = 2, 3$.

The algebra induced by E on V_i/A_i is explicitly obtained by restricting its basis matrices to the quotient; denote it by E_i. The ideal $I_\tau(E)$ acts as zero there. Hence E_i is an image of $E/I_\tau(E)$, and

$$\dim_{\mathbb{F}} B \le 100 r^2 \quad \text{for every simple factor } B \text{ of } E_i/J(E_i). \tag{7.8}$$

Here are the rank and group bounds that will be used together. Let U_i be a copy of A_i, with the specified inclusion $\iota_i : U_i \to V_i$. On the three spaces $V_i \oplus U_i$, put

$$\begin{aligned} F_T((v_1,u_1),(v_2,u_2),(v_3,u_3)) = {} & T(\iota_1 u_1, v_2, v_3) + T(v_1, \iota_2 u_2, v_3) \\ & + T(v_1, v_2, \iota_3 u_3). \end{aligned} \tag{7.9}$$

Each summand is a pullback of one R_i, so (7.7) and subadditivity imply

$$\mathrm{ark}(F_T) \le 3\alpha. \tag{7.10}$$

Prescribing the summands V_i, U_i and the graph $\{(\iota_i u, u) : u \in U_i\}$ forces an allowed map on $V_i \oplus U_i$ to be the direct sum of a map on V_i and its restriction to A_i. Thus matching (7.9) is equivalent to matching T modulo

$$\mathcal{Q} = \operatorname{ann}(A_1) \otimes \operatorname{ann}(A_2) \otimes \operatorname{ann}(A_3) \leq V_1^* \otimes V_2^* \otimes V_3^*. \tag{7.11}$$

Indeed a trilinear form belongs to $\mathcal{Q}$ exactly when it vanishes whenever one of its inputs belongs to the corresponding A_i.

Let H be the full stabilizer of $T + \mathcal{Q}$, the A_i, and all the ordered forms Q_{h_j}, within $\prod_i \mathrm{GL}(V_i)$. If $h \in H$, then $h^{\mathsf{t}} Q_{h_j} h = Q_{h_j}$, so $(h, h^{-1}) \in E$. The induced maps of h on the quotient factors belong to $E_i^{\times}$. This proves that the group returned by the constrained low analytic rank algorithm satisfies the algebra hypothesis of Proposition 5.2, with the bound in (7.8).

7.4 Enumerating target images and completing an isomorphism

The target enumeration consists of all ordered lists $h'_1, \ldots, h'_t \in U' = V'_1 \oplus V'_2 \oplus V'_3$. Its size is $q^{\ell t}$. For each list, form $Q'_{h'_j}$ from T', construct its adjoint algebra E', and put $A'_i = I_\tau(E')V'_i$. If $\dim A'_i \neq \dim A_i$ for some i, discard this list.

For the remaining list apply Proposition 5.2 to T, T', the subspaces A_i, A'_i, and the ordered bilinear forms comprising $Q_{h_j}, Q'_{h'_j}$. Its hypotheses follow from (7.7) and (7.8). We describe its two steps here to specify the actual decision performed on a target list.

First, construct $F_T, F_{T'}$ and the graph constraints above. Theorem 3.1 either finds their full constrained transporter $g_0 H$, expressed on the original spaces, or decides that it is empty. The former transporter consists exactly of the maps preserving the ordered forms and taking $T + \mathcal{Q}$ to $T' + \mathcal{Q}'$. By (7.10), this step takes

$$q^{O(N+\alpha^2)} \operatorname{poly}(t, \log q)$$

time. The bound in Theorem 3.1 uses the analytic rank of F_T; it therefore applies to every enumerated target list.

When this transporter is nonempty, pull T' back by g_0 and put $y = g_0^* T' - T \in \mathcal{Q}$. Use the left action on forms

$$\rho(h) = h_1^{-\mathsf{t}} \otimes h_2^{-\mathsf{t}} \otimes h_3^{-\mathsf{t}}, \qquad c(h) = \rho(h)T - T \in \mathcal{Q}.$$

It satisfies $c(hk) = c(h) + \rho(h)c(k)$. Thus $h \cdot z = \rho(h)z + c(h)$ is an affine action of H on $\mathcal{Q}$. Its linear part on $\mathcal{Q}$ is the tensor product of the inverse transpose quotient actions. The opposite algebras obtained by transposing the E_i have the same simple factor dimensions. Theorem 4.4 decides whether

$$h \cdot 0 = y \tag{7.12}$$

for some $h \in H$ in $q^{O(N+100r^2)} \operatorname{poly}(t, \log q)$ time. If it returns such an h, then $\rho(h)T = g_0^* T'$, and pulling back by h gives $(g_0 h)^* T' = T$. Return $g_0 h$. If no target list succeeds, return that the tensors are not isomorphic. An accepted triple may also be checked directly using (1.1), in polynomial time.

7.5 Correctness and running time

Every returned triple satisfies (1.1), by (7.12). For the converse, suppose $g = (g_1, g_2, g_3)$ is an isomorphism from T to T'. The target enumeration contains the list $h'_j = g h_j$. Multilinearity gives

$$g^{\mathsf{t}} Q'_{g h_j} g = Q_{h_j} \qquad (j \in [t]).$$

On adjoint pairs, the map $(a, b) \mapsto (gag^{-1}, gbg^{-1})$ identifies E and E'. The faithful representations are intertwined by $\mathrm{diag}(g, g^{-\mathsf{t}})$ on W. Since the ideal I_τ is preserved by algebra isomorphisms, this map takes $I_\tau(E)W$ to $I_\tau(E')W'$ and A_i to A'_i. The target list therefore passes the dimension comparisons.

The map g belongs to the constrained quotient transporter. Hence the first step returns a nonempty coset g_0H containing g. Write $g = g_0h$. The identity $g^*T' = T$ gives exactly (7.12), so the affine orbit step succeeds on this list. This proves both directions of correctness. The source choices are made only once; the enumeration of all their target images supplies the list just used in this argument.

By (7.4), the target enumeration costs $q^{O(N^2 \log(N+2)/r)}$. Computing the target algebra data and performing the two subsequent steps for one list costs

$$q^{O(N+\alpha^2+r^2)} \operatorname{poly}(t, \log q) = q^{O(N+r^2+N^4/r^4)} \operatorname{poly}(t, \log q).$$

The source computation costs $q^{O(N)} \operatorname{poly}(\log q)$. Polynomial factors in N and t are absorbed into $q^{O(N)}$. The total is therefore

$$q^{O(N^2 \log(N+2)/r+N+r^2+N^4/r^4)} \operatorname{poly}(\log q). \tag{7.13}$$

Choose r to be the least positive integer with $r^3 \geq N^2$, namely $r = \lceil N^{2/3} \rceil$. Then $r^2 = O(N^{4/3})$ and $N^4/r^4 = O(N^{4/3})$. Substitution in (7.13), together with $N = 2\ell$, proves Theorem 1.1.

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